

ABSTRACT

DESIGN AND CONSTRUCTION OF TAPERED AMPLIFIER SYSTEMS FOR LASER COOLING AND ATOM TRAPPING EXPERIMENTS

by Jayampathi Kangara

We present detailed plans for the construction and operation of a tapered amplifier (TA) system seeded by a single-mode, frequency-tunable near-IR external cavity diode laser. Our plans include machine drawings for the parts, electronic circuit diagrams, and information on prices and vendors. Instructions are provided on how to safely couple light into and out of the TA chip, and other practical aspects of handling the chip are discussed. Many cold atom experiments require light beams with Gaussian spatial profiles - measurements of the tapered amplifier light output through a single-mode optical fiber are presented as a function of seed intensity and driving current.

DESIGN AND CONSTRUCTION OF TAPERED AMPLIFIER SYSTEMS
FOR LASER COOLING AND ATOM TRAPPING EXPERIMENTS

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Dedication

To my wonderful wife, Harshani

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Finally I'd like to thank Diane Beamer for helping us with some of the figures given in this thesis.

Chapter I - Introduction

I.1. Motivation

The ever-growing presence in the advanced undergraduate laboratory of the frequency-tunable external cavity diode laser (ECDL), operating in single transverse and longitudinal mode with a linewidth of 1MHz, is well documented[1],[2],[3] These lasers have enabled the advent into the undergraduate experimental curriculum of interesting topics in atomic and optical physics such as laser cooling and atom trapping [4],[5],[6],[7], electromagnetically induced transparency[8],[9], and novel techniques in Doppler-free and sub-natural linewidth spectroscopy [10],[11],[12].

It is well known that while ECDL's offer important advantages such as low cost, narrow linewidth, and wide tunability, they are limited in output power. After beam- shaping, Faraday isolation, and fiber-coupling, one is typically left with less than ten mW [13],[14], although up to several tens of mW may be obtained by injecting the output into another slave" diode laser [15],[16]. This is because the requirement for the light output to be single spatial mode forces the transverse dimension of the diode laser used in the ECDL to be on the order of the optical wavelength. If the goal is to attain $\geq 1\text{W}$ of optical power suitable for state-of-the-art experimentation in laser cooling and atom trapping, a popular method is to use a tapered amplifier (TA) chip - a semiconductor device which achieves high power while retaining the narrow linewidth and stability of the ECDL[17]. TA devices are available commercially in a price range \$14,000 - \$25,000 per unit, depending on the choice of maximum output power (0.15W - 2.5W), center wavelength (650 nm - 1190 nm), and desired accessories (for instance, Faraday isolation and/or fiber coupling in to and out of the chip).

In this thesis, we present detailed plans for the design and construction of a TA system seeded by a continuous wave ECDL. We combine features from previous designs in important aspects such as mode coupling and current/temperature regulation, enabling the demonstration of a stable, compact, inexpensive TA system for laser cooling and atom trapping experiments at 780 nm. Measurements of the TA output after collimation, as well as after passage through a single-mode optical fiber, are presented as a function of driving current and seed laser intensity. Since TA chips are notoriously susceptible to permanent damage by even small amounts of optical retro reflection and voltage fluctuation, various safety precautions are described to ensure that the TA system is suitable for implementation in undergraduate laboratories.

I.2. Background on Tapered Amplifier Systems, and Previous Work

TA chips are notoriously fickle, susceptible to permanent damage by voltage spikes and the slightest of optical retro-reflections. Over the past few years several research groups have proposed detailed designs for damage free TA systems[17][16]. The safety mechanisms and design characteristics are different in different systems.

When discussing the thermal requirements for the safe operation of the TA chip it is important to realize that the actual TA chip is just 2mm x 205 μ m. When the chip is working, this small chip generates a huge amount of optical power (at 1.5 A for 15 mW seed, the chip puts out ~450 mW of optical power), and also it is pumped by > 1 A current. So the chip generates a huge amount of heat while operating. If there is no mechanism to cool the chip (by allowing the heat to dissipate) the chip will not last even a minute. In Nyman's design [17] they use two Cu blocks to mount the chip and a water cooled Cu base plate along with a thermoelectric cooler, or TEC [17]. But in Fuch's [16] and in Yavuz's [18] designs they bypass water cooling methods (undesirable because of the constant threat of a leak) and instead use the two Cu blocks (to mount the chip) in conjunction with a large Al block which act as good heat sinks to extract the heat from the chip [16][18] without the need for any water cooling thus simplifying the setup.

Another important thing is that to get the maximum power output from the chip the coupling of the input beam in to the amplifier chip should be almost perfect. Because of the small aperture size of the input facet of the chip (1-3 μ m), a tightly focusing lens is used to focus the incoming beam in to the chip. The output power strongly depends on the position of this lens. But in Nyman's design [17] they have attached this lens permanently, down to the Cu mount on which the chip sits [17]. So it is not possible to adjust this lens while the chip is working. In Fuch's design they've glued this lens to two posts and then to a translational stage[16] – however, they mention that even slight disturbances such as blowing against the translational stage and leaning on the optical table can change the output of the laser[16]. In Yavuz's design they have solved this problem by mounting the lens inside a custom built lens tube which allows the positioning of the lens to a few micron precision and also provide better mechanical stability [18]. In our design, described in the following section, we have combined the best-design features from various leading groups, and put in several electronic and optical safeguards to come up with a user-friendly and relatively fail-safe design.

Chapter II – Overview of Components

II.1. TA Assembly

The complete assembly for the TA system is shown in Figure 1. The chip itself is located between two copper blocks that house collimation lenses on the input and output sides of the chip. The copper blocks sit atop a copper base plate in which a groove is cut to accommodate a thermistor. To the bottom of this plate is attached a ceramic thermoelectric cooler (TEC) device, which itself sits on top of a large aluminum heat sink. Holes and passageways are cut into the copper and aluminum pieces to allow for electrical wires to pass through for the purposes of supplying current to the TA diode and for temperature regulation. In this section we go over each of the basic components of the TA system.

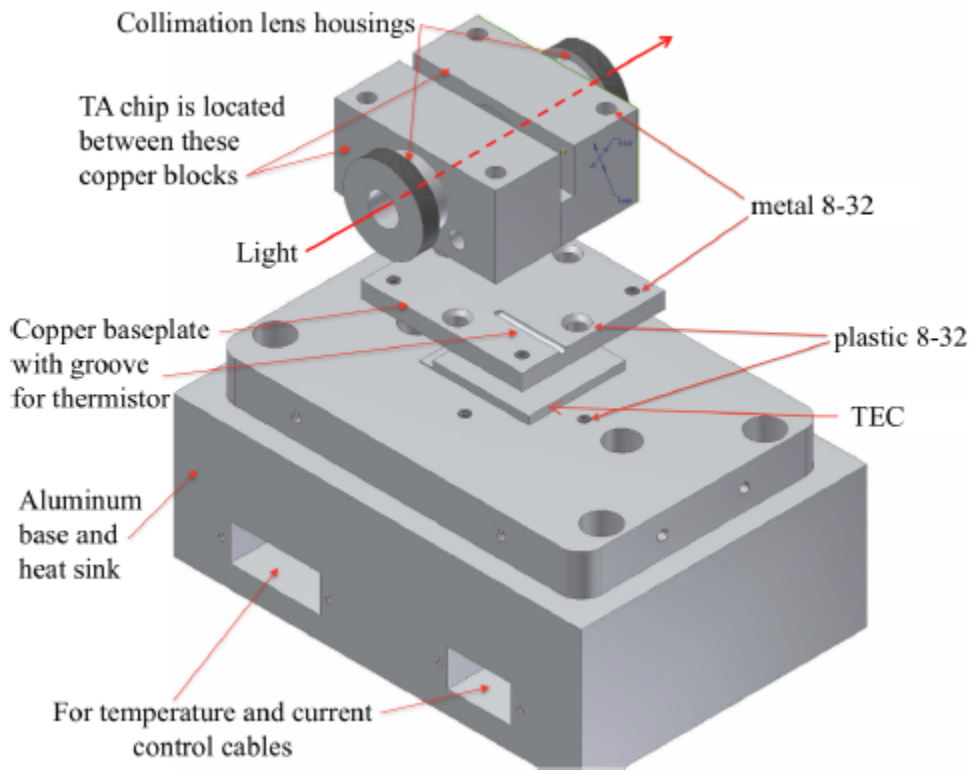


Figure 1: View of the complete TA system after assembly. Reproduced from Ref. [19].

II.2. The TA chip

We have used TA chips made by m2k-Laser GmbH (Model m2k-TA-0780-1000-CM, rated for 1W of continuous wave power in the wavelength range 767 - 787 nm, 765 - 790 nm in more recent versions), which we found to be substantially less expensive than the more commonly used chips made by Eagleyard. A picture of this chip, on a C-mount, is shown in Figure 2.

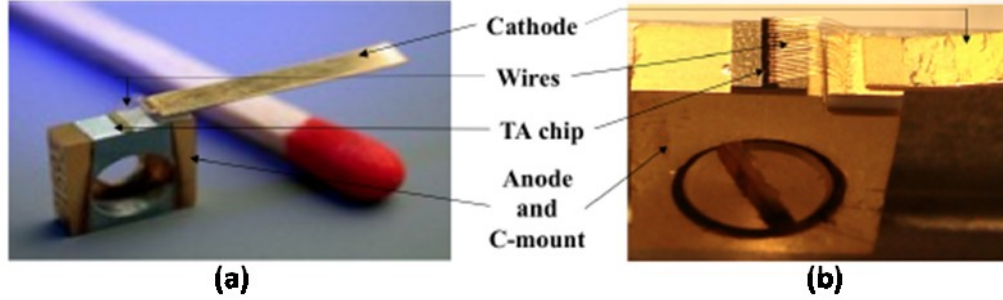


Figure 2: a) The TA chip comes attached to a C-mount which also serves as the anode. The cathode is in the form of a winged appendage. b) A close-up photograph of our chip showing the wires used to electrically pump the tapered region of the chip

The basic structure of the TA chip has been described in detail before [16],[20],[17],[21]. A simplified working model, for the purpose of this work, is depicted in Figure 3. In this highly simplified picture, the TA chip consists of a short straight index-guided waveguide section ($L_1 \sim 0.5$ mm typically) which is coupled to a longer gain-guided tapered section ($L_2 \sim 1.5 - 3$ mm; in our specific case, $L_1 + L_2 = 2$ mm, though for the newer version of this chip the value is 2.5 mm). Light is injected through the narrow aperture (typically, $W_1 \sim 1-3$ μm ; for our chip $W_1 = 1.3$ μm , and 1.2 μm in the updated version of our chip) of the index-guided section. The amplified output is emitted through the much wider aperture (typically, $W_2 \sim 200$ μm ; 205 μm for our specific chip, 150 μm in the latest version) of the tapered gain-guided section, as shown in Figure 3.

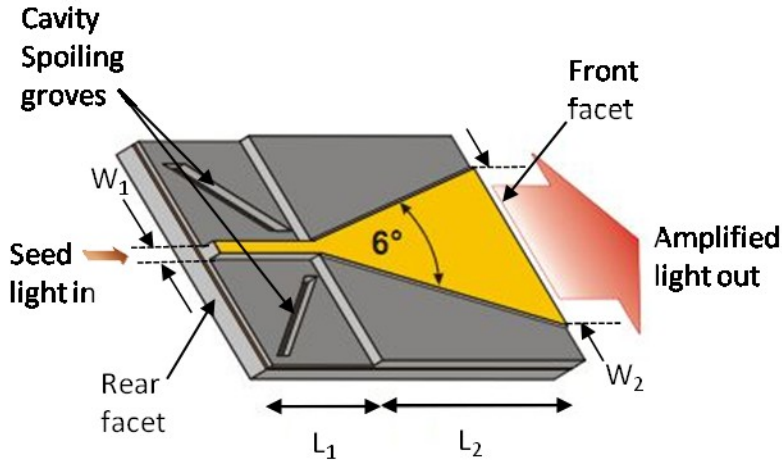


Figure 3: Schematic of a TA chip with the ridge waveguide and the tapered gain region

The taper angle is $\sim 6^\circ$, chosen such that the beam from the straight section diffracts and expands to fill the entire tapered region. In Figure 2(b) note the presence of closely spaced wires in order to provide uniform electrical pumping to the tapered section of the chip. Since the gain varies with the inverse of the optical power density the incoming Gaussian beam experiences a higher gain towards the ‘wings’ and a lower gain at the center [16]. Ideally this leads to a good “flat-hat” distribution for the out coming beam profile.

The electrical contact to the chip is established by metalizing the top and bottom surfaces of the chip, thus connecting the top surface to the cathode wing and the bottom surface to the C-mount which serves as the anode (See Figure 2 (b)).

The small transverse dimensions of the straight section ensure that only the fundamental transverse mode is excited, yet this section by itself would restrict the user to low output powers for a given energy density. Coupling to the broader gain region of the tapered section enables the production of high power for the same value of the energy density[16]. The purpose of etching cavity-spoiling grooves into the chip is to deflect and scatter unwanted oscillating modes: Such modes propagate with wave-fronts parallel to the two facets (also known as Fabry-Perot cavity modes) outside the tapered region. If allowed to oscillate these modes would optically pump regions of the chip that are not electrically pumped, thus heating up the chip [20].

II.2. Collimating lenses and lens mount assembly

An aspherical lens (Thorlabs C230TME-B, focal length= 4.51 mm) is placed on either side of the TA chip, one to focus the seed beam from the ECDL into the tiny TA chip, and the other to collimate the highly divergent output from the TA. The lenses are housed inside custom- made mounting assemblies which are, in turn, housed inside the copper blocks seen in Fig. 1. These blocks are made out of oxygen-free high-conductivity copper because the anode of the TA chip is in direct electrical and thermal contact with them. The C-mount of the TA chip is screwed into a slot machined in one of the blocks, as shown in Figure 4. The slot dimensions are such that the TA chip snugly fits inside and the gain medium of the chip is centered on the optical axis of the collimation lenses.

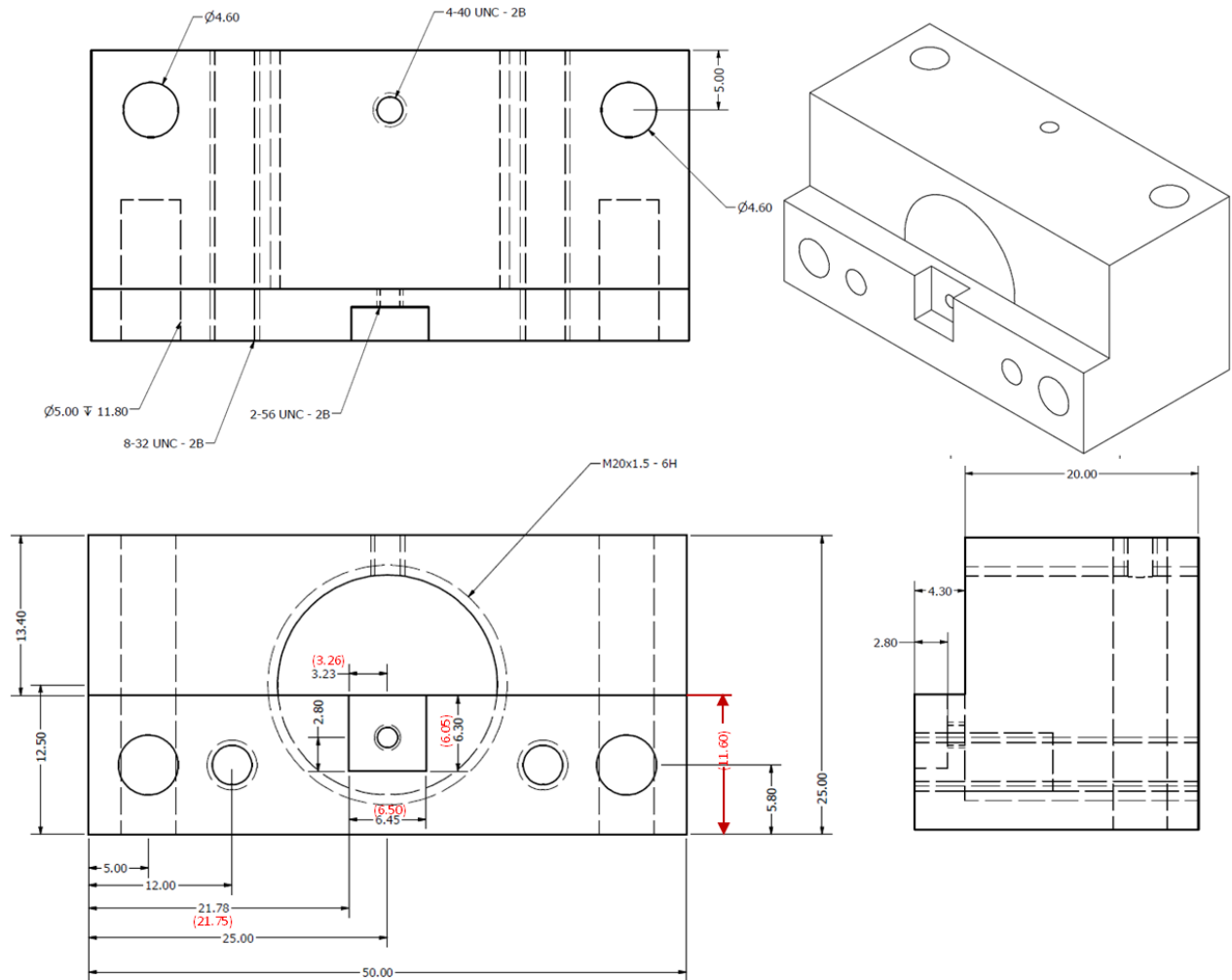


Figure 4: Machine drawings for the copper block that contains a slot for the TA chip. The numbers in black are for the Eagleyard chip EYP-TPA-0780-01000 for which these drawings are originally made. The numbers in red are modifications for the m2k-laser chip used in our lab.

The drawings for the other block are shown in Figure 5.

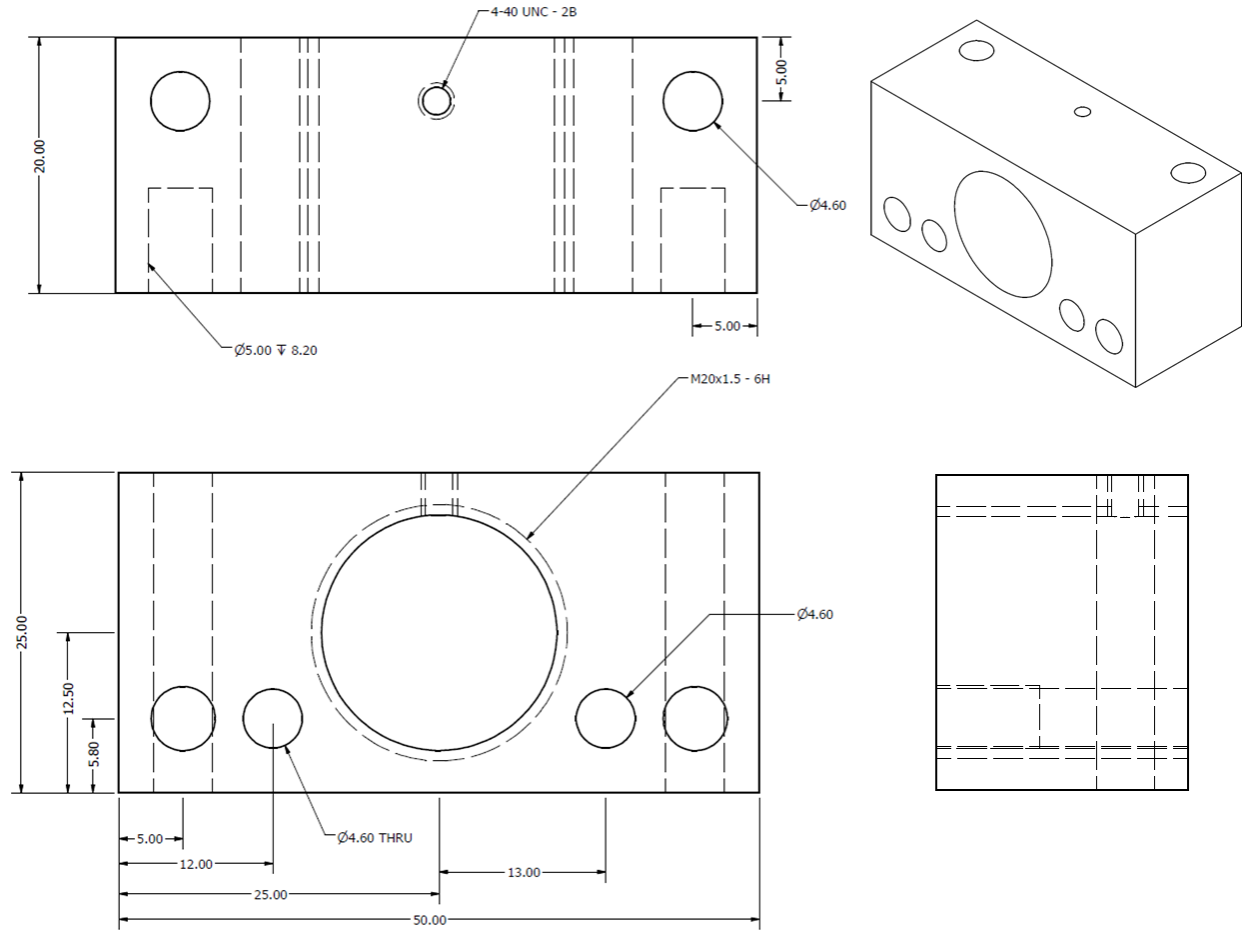


Figure 5: Machine drawings for the copper block facing the one that contains the TA chip.

Figure 6 shows a picture of the completed copper blocks. Even though both these Cu blocks are screwed on to the small Cu base and to the rest of the system it is important to realize that there can be a certain misalignments of these Cu blocks on the horizontal plane. This is a crucial factor because it can change the focal point of the lenses on the parallel plane to the optical table. Since our lens tubes are not capable of moving on this parallel plane it is important to align the two Cu blocks initially in order to get the central axes of both the Cu blocks on to single axis which lies along the center of the TA chip. Both copper blocks have centrally located tapped holes to enable the introduction of the collimating lens and their mount assemblies as discussed in the following paragraphs.

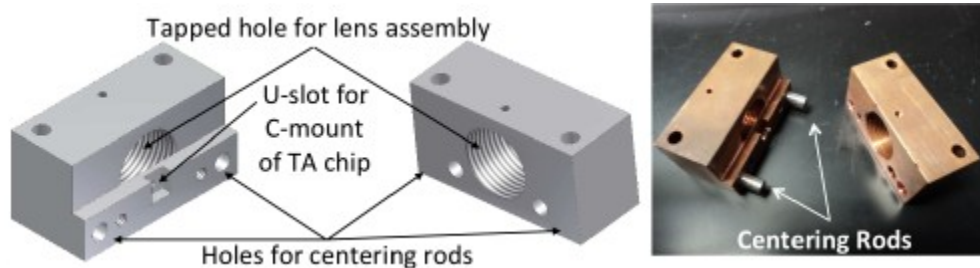


Figure 6: The completed copper blocks for housing the TA chip and collimating lens assemblies. The centering rods for the alignment of the blocks are shown. Note: The tapped hole on the top of the right hand piece is in the wrong position

Since the incoming beam should be focused tightly in to the TA input facet, an aspheric lens ($f=4.51$ mm, 0.55NA, ThorLabs C230TME-B) is used. Even on the output side of the chip, we used the same type of lens to collimate the outgoing beam. These lenses are mounted inside two lens tubes which have been screwed in to the Cu blocks on the input and output sides of the chip. Outer cylinder (thread: M20x1.5 – 6g) is screwed in to the Cu block while the inner cylinder, where the actual aspheric lens sits in, is screwed in to the outer cylinder. This inner cylinder consists of finely made thread (thread: M16x0.5 – 6H), allowing a smooth movement of the Aspheric lens with respect to the position of the TA chip. Each copper block has an identical inner and outer lens assembly.

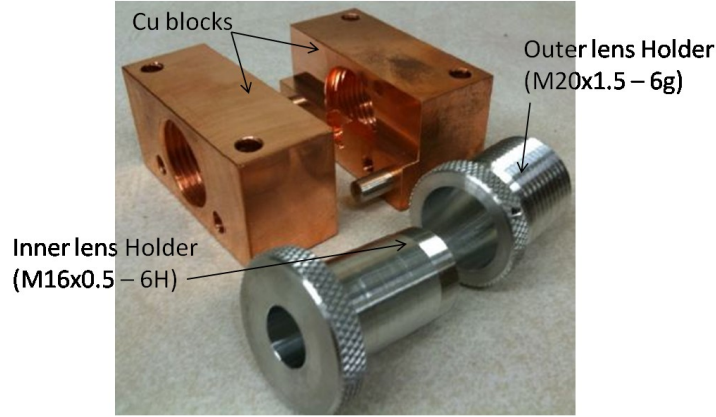


Figure 7: Lens mounts and the two Cu blocks

Aspheric lens is screwed in to a mating thread inside the inner lens tube. A set screw in the copper block locks the outer lens cylinder after the coarse alignment of the aspheric lens and a set screw in the knurled grip of the outer lens cylinder holds the inner lens cylinder in place after the fine adjustment of the collimation. Technical drawings for these two lens holders are given in Figure 8. A knurled grip for the inner lens mount is not shown in the technical diagram.

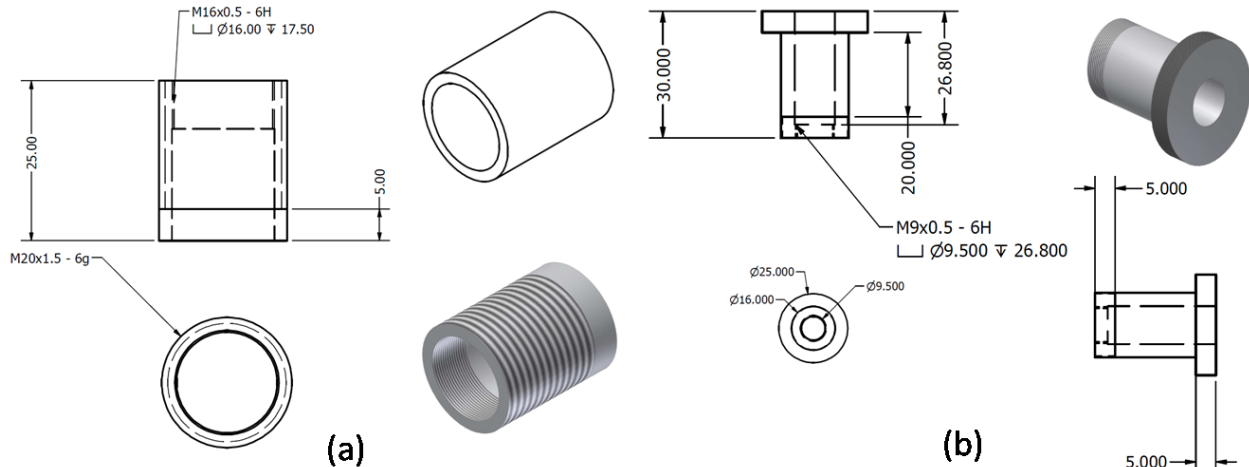


Figure 8: a) Technical diagram for the outer cylinder. b) Technical diagram for the inner cylinder

II.3. Thermoelectric cooler, Thermistor, Heat sink

Temperature regulation of the TA chip is of critical importance because during the operation the chip generates a huge amount of optical power ($\leq 1W$) while being electrically pumped by currents between 1.5A to 2.5A. The lifetime of the chip has a strong dependence on the operating temperature of the chip[16]. It is believed that operating at high temperatures ($>30^0$) can reduce the lifetime of the chip. On the other hand, operating around lower temperatures ($<15^0C$) can cause water condensation inside the housing of the chip. As seen in Figure 1 the temperature is monitored by a thermistor (Model TCS10K5 from Wavelength Electronics) affixed with heat sink paste inside a groove cut into a copper plate that serves as a base plate for the copper mounts housing the chip and the collimation lens assembly. A mechanical drawing for this copper base plate is given in Figure 9.

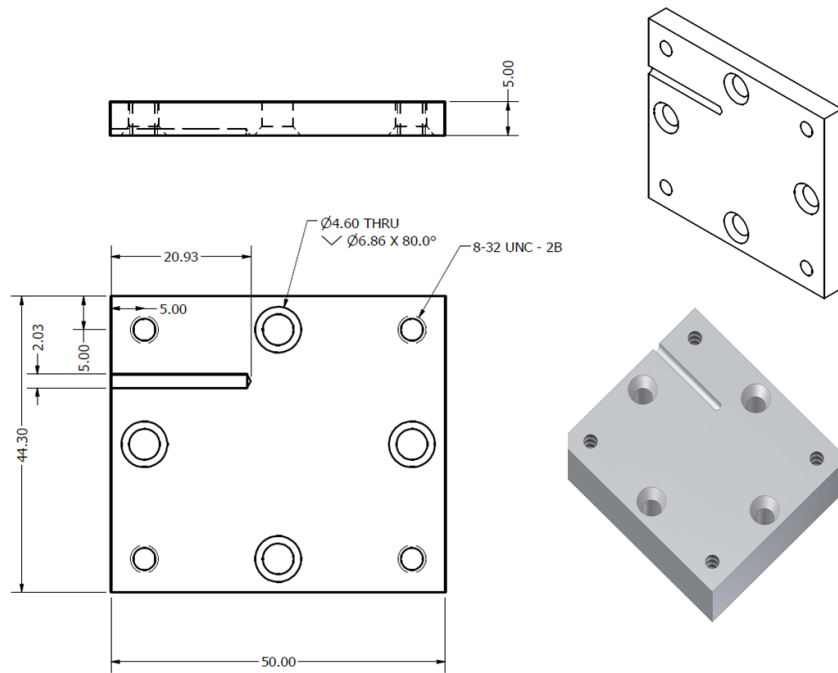


Figure 9: The copper base plate with a groove for the thermistor

As Figure 1 shows temperature regulation is provided by a thermoelectric cooler (TEC; Model CP14,71,045 from Laird Technologies), inserted between the copper base plate shown in Figure 9 and a large aluminum base. In a previous design, water cooling of the copper blocks was required [17], but in this design the aluminum base (see Figure 10) acts as an adequate heat sink without the need for water cooling [16],[19]. A machine drawing for the aluminum base is shown in Figure 10. Note from Figure 1 that the slots cut in to allow plugging in of cables for the circuits providing current to the TA chip and to the TEC are on the input side, not on the output side where the cables would obstruct the placement of a cylindrical lens placed immediately after the TA system (use of the cylindrical lens is described in the upcoming section).

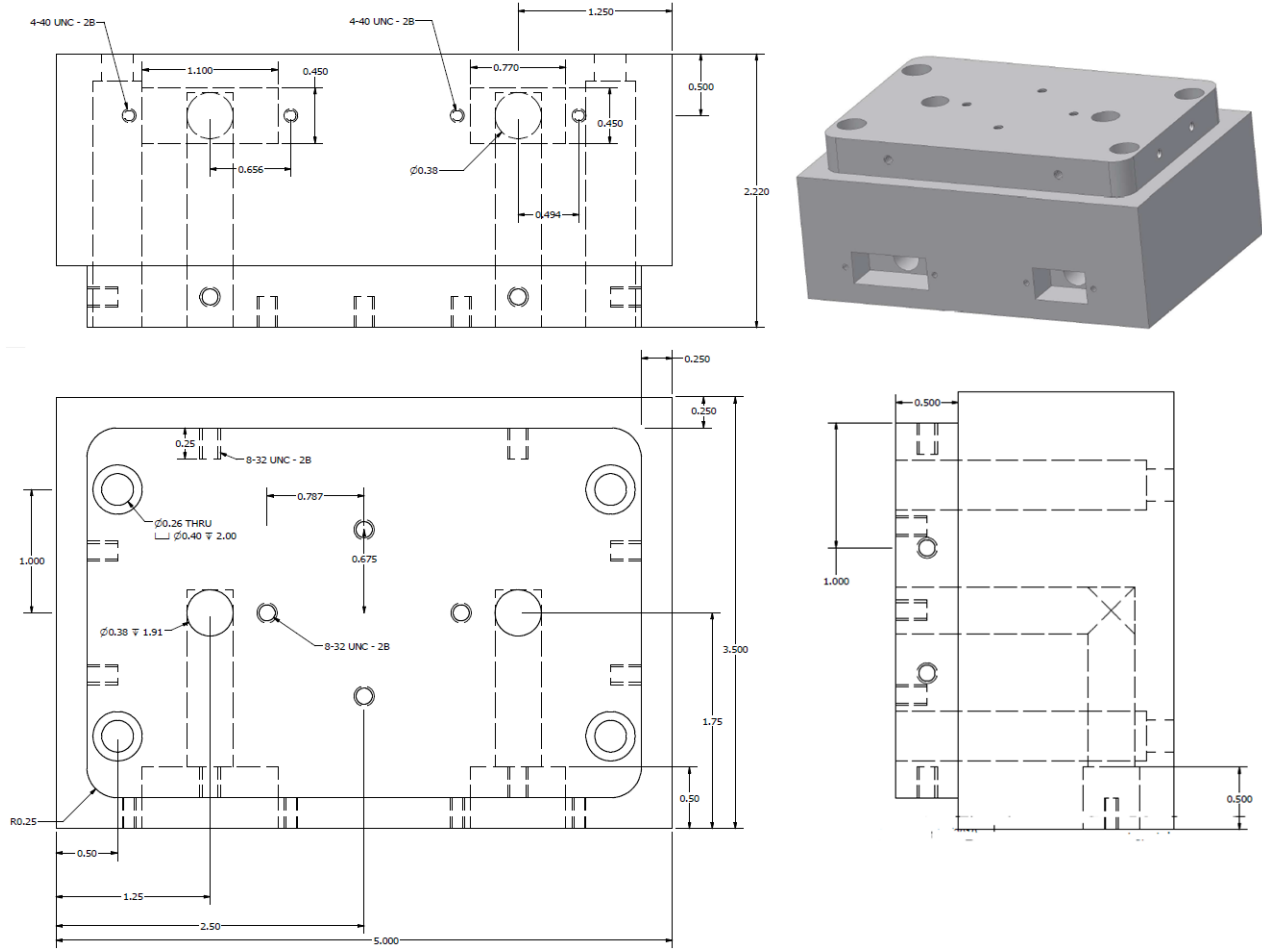


Figure 10: The aluminum base plate which serves as a heat sink. The top right hand drawing is meant to give an overall view and is not to scale with other drawings in this figure

II.4. Cylindrical lens

It is well-known that the output beam from the TA chip is not only highly divergent but also highly astigmatic [16, 20, 17]. The beam divergences are very different in each of the perpendicular transverse directions, in our case 6° in the horizontal direction and 45° in the vertical direction (see Figure 11) leading to different focal points for the two perpendicular directions after passing through a spherical converging lens. The aspherical lens in the previous sub section serves to collimate the output beam in the vertical direction. An antireflection coated cylindrical lens (Thorlabs LJ1821L1-B, focal length = 50mm) placed just outside the plexi-glass housing for the TA system is used to collimate the beam in the horizontal direction. One should be careful to slightly angle the cylindrical lens so that any retro reflected light does not get sent back in to the TA chip.

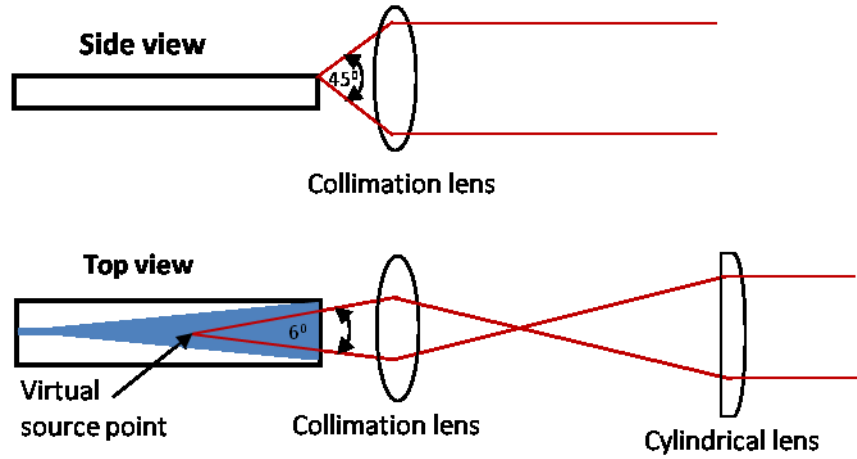


Figure 11: Top and Side view of the output side of the chip with the outgoing beam showing the Aspheric and the cylindrical lenses. Note that in the plane of the chip the beam diverges from a virtual source that is approximately L_2/n_L behind the output facet (where L_2 is the length of the gain region and n_L is the effective refractive index[16]).

Chapter III - Assembly of components

III.1. Handling the TA chip

Usual precautions, similar to those observed while handling laser diodes, such as the operator needing to be electrically grounded before handling the mounted chip, need to be carefully followed. The electrical circuit for supplying the current to the TA diode is described in section III.3. The current driver includes a protection circuit which is connected across the terminal of the diode (more details about this protection circuit is given in section III.3. The wire going from the protection circuit to the cathode needs to be carefully soldered to the metal wing protruding from the TA chip. We found this easier to do while the chip was still screwed in to its original mount in the box supplied by the manufacture. After doing this the chip may now be removed from the box- we found it simplest to use a pair of small needle-nosed pliers to grab the C-mount and hold the wire attached to the cathode wing for light positioning- and settled into the U-shaped slot in the copper lens mount in the Figure 6 . It is, of course, imperative to make sure the input and the output sides of the chip are correctly identified (see Figure 12), for injecting the seed beam through the wrong end would instantly damage the chip.

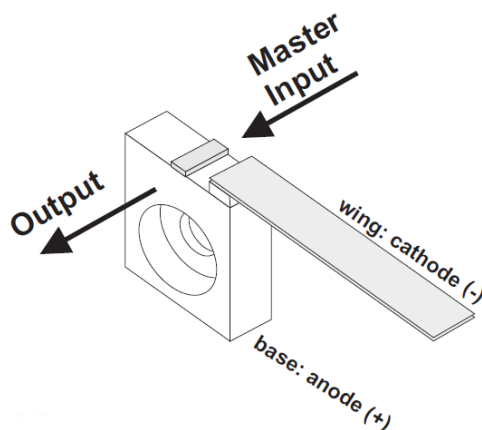


Figure 12: Input and Output sides of the TA chip

Equally important extreme caution must be taken to never, under any circumstances, let anything touch the delicate wires shown in Figure 13 which bond to the chip. Figure 13 shows a close-up image of damage to wires bonded to the cathode owing to slight inadvertent mechanical contact by a user, which resulted in failure of the diode.

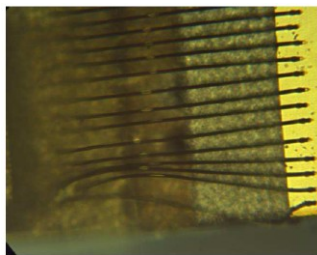


Figure 13: Picture of a chip with damage to the wires bonded to the cathode.

The wire from the protection circuit for the anode is simply soldered to a grounding lug which is attached to an 8-32 screw that goes through a clearance hole in the lens mount in Figure 4 before screwing in to the tapped hole in the base-plate in Figure 9. As is clear from Figure 1, Figure 4 and Figure 9 the base plate and lens are in direct electrical contact with C-mount (anode) for the TA chip.

III. 2. Placing the TEC

Thermally conductive heat paste is used between at both TEC-aluminum interfaces. It is crucial when putting these parts together to not use too much heat paste. If the interfaces between the parts are smooth and contact is good, one may get away without using any heat paste at all. If one does use heat paste, a good rule is to use less than half a grain of rice. The reason is that heat paste is not all that thermally conductive. It is not helpful in conducting heat through from one part to another if the interface between the two parts is physically rough, and there are lots of tiny or microscopic gaps where there is no contact.

III.3. Design of the TA current and temperature controller

The TA current and temperature controllers were designed by the Instrumentation Laboratory according to specifications provided by us. The Temperature controller is capable of reading the actual temperature of the TA assembly and locking the temperature to a certain adjustable set point value. The Temperature controller follows the design of a usual temperature locking circuit which includes a PID loop. But there is something interesting about our current controller which distinguishes it from a usual laser diode current controller. Our current controller includes several important safety mechanisms which are essential for reliable operation of our tapered amplifier chip. In the following sections I will be talking about each of these mechanisms separately.

III.4. Collimating the input side of the TA

Amplified Spontaneous Emission (ASE) at the input side of the chip should be collimated first. At this point one should block the seed beam and must look at the light coming out from the back facet of the TA with an IR viewer. It is important to keep in mind that the current of the TA should be kept at a lower value (< 750 mA) in order to prevent any unwanted heating effects which could lead to a degradation of the chip. Collimation of the ASE should be done as perfectly as possible because the output power depends highly on the position of the lens with respect to the input facet of the chip. The seed is then aligned so as to trace the path of the collimated ASE, in to TA chip. More details about how to inject the seed in to the TA for the first time will be discussing under section IV.

Chapter IV- Operation of the TA system

IV.1. Optical setup on the input side of the TA

Figure 14 below shows the optical layout on the input side for the TA chip. An external cavity diode laser (ECDL) built in a Littrow Configuration has been used as the seed of the TA. When the beam comes out from the ECDL it is elliptical due to the shape of the output facet of the diode laser. To make the beam circular an anamorphic prism pair has been used. A circular beam can be obtained by changing the relative angle of the prism pairs. After that an optical Isolator has been placed to avoid the back reflection to the laser. This is important because when the TA chip is operating it puts out a secondary beam from the rear facet of the chip due to the amplified spontaneous emission (ASE) inside the gain medium. If this beam goes back to the seed laser it can destroy the single mode behavior of the laser and at the same time it causes lot of distortions on the seed laser making it hard to lock the frequency for atom trapping experiments. Appropriate polarization for the Isolator is achieved by using a half-wave plate. After the isolator the beam goes through an electronic shutter, which I will be discussing in more detail on section IV.1.A. Then we have a half-wave plate followed by a polarizing beam splitter which reflects a certain portion of the beam to a photo diode of the second protection circuit described in Section IV.1.B to lower the TA current in case there is a sudden shutdown of the seed. Finally the beam goes through two mirrors and a half-wave plate before it enters the TA. The last two mirrors have been placed there because then it becomes possible to align the beam through the TA. The half-wave plate is there because the gain of the chip is highly sensitive to the incoming polarization [16].

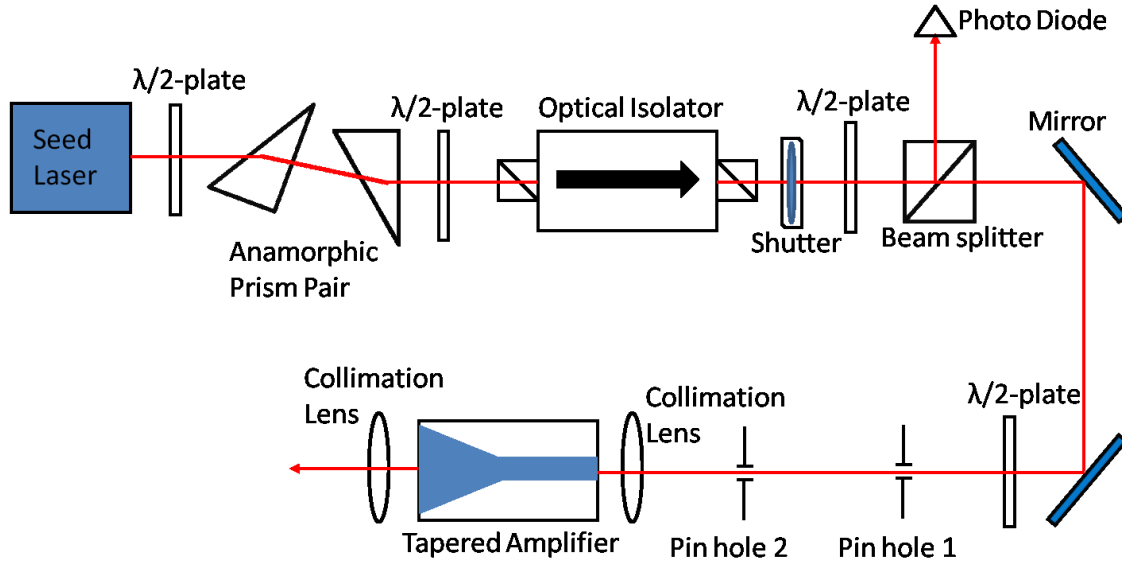


Figure 14: Optical setup for the input side of the TA

IV.1.A. Design of the shutter mechanism to prevent seed input while chip is powered OFF:

For the TA chip, it is important to not have a seed going in while the chip is turned OFF. This is achieved by using an external shutter. The current controller was modified in order to put out a +5V step voltage after it passes a certain current limit (120 mA-For the Trap TA). Then a Shutter

controller circuit was designed to operate a shutter to block and unblock the beam according to the pulse, put out by the TA current controller. Again due the higher prices of commercial shutter controller systems, we had to buy each component separately (shutter and power supply for the shutter driver) and make the Instrumentation lab assemble all the components together in to a complete system. The shutter controller, when connected to the TA controller, takes the signal from the TA shutter output (+5V step voltage) and sends another pulse to the shutter to OPEN or to CLOSE it. So if we put the shutter in the path of the seed laser it act as a gate for the input for the TA chip.

IV.1.B. Design of a Protection mechanism for the situations where the seed gets blocked accidentally:

Operating the TA with high currents for long time periods without having the seed laser can be a reason for the chip to degrade eventually [16]. So we do not want to operate the TA chip at higher current values when the seed is not present. If the seed laser gets accidentally blocked we want the current of TA chip to go down to a safe limit. We achieved this by putting a polarizing beam splitter in the path of the seed and guiding a certain portion of the beam in to a monitor photodiode. If something happens to the seed and if it gets blocked accidentally, the TA controller recognizes this change by looking at the photodiode voltage and decreases the TA current to a safe limit.

Our current controller is designed for minimum fluctuations in the output. But for further protection, we asked the Instrumentation Lab to design another small protection circuit which sits inside TA box. This circuit consists of diodes and capacitors – the diodes to prevent voltage fluctuations of the wrong polarity to appear between the anode and cathode of the TA chip, and the capacitors to smoothen out sharp voltage spikes that may damage the chip. Circuit diagram for this circuit is given on page 45.

IV.2. Injecting the seed in to the TA for the first time

There are several things one has to do when you first couple the beam in to the TA. First the Amplified Spontaneous Emission (ASE) the back facet should be collimated. At this point one should block the seed beam and must look at the light coming out from the back facet of the TA with an IR viewer. It is important to keep in mind that the current of the TA should be kept at a lower value (< 750 mA) in order to prevent any unwanted heating effects which could lead to a degradation of the chip. Collimation of the ASE should be done as perfectly as possible because the output power depends highly on the position of the lens with respect to the input facet of the chip. The seed is then aligned so as to trace the path of the collimated ASE, into the TA chip. To do this, we followed the below mentioned procedure.

After collimating the ASE on the input side, set up two pin holes approximately about 300mm away from each other marking the path of the ASE light. These two should be setup in between the second coupling mirror and the input side of the TA (see Figure 14). Then unblock the seed and use the coupling mirrors to guide the beam through the two pinholes to overlap the ASE and the seed on top of each other. After getting the coarse adjustments, open up the both pinholes and let the seed go through the TA chip. At this point one must have the power meter setup on the output side of the TA to measure the output power. Block and unblock the seed to see if there is

a change in the output power with the seed beam going in. If you don't see a change then go back to the two pin holes and check the alignment of the seed again. Even if a slightest amount of the seed beam is going in to the TA you must see a change in the output power when you block and unblock the seed (it's really IMPORTANT to keep the TA current below 750 mA when you follow this procedure). If you see a change, then further effort should be taken out to improve the coupling efficiency by walking between the two coupling mirrors.

Output power strongly depends on the position of the input collimation lens of the TA. So after adjusting the coupling mirrors several times one might also need to touch the input lens tube to improve the power further. This needs to be done very carefully and since we've already adjusted the lens when collimating the ASE, only a slight adjustment is needed to get it finely aligned. Again one should be wearing the grounding strap when doing this step.

We also found that whenever we adjust the collimation of the seeding laser we are required to touch the collimation lens on the TA and the two coupling mirrors to re-optimize the seed back in to the TA chip.

IV.3. Optical setup on the output side

When the amplified beam comes out from the output facet of the TA it diverges rapidly in both vertical and horizontal directions. As mentioned in Section I.2 the beam shows astigmatism. This means one should use two lenses to collimate the beam in both horizontal and vertical directions. Vertical direction is corrected by the same type of aspheric lens ($f=4.51$ mm, 0.55NA, ThorLabs C230TME-B) which we have used on the input facet while the horizontal direction is corrected using a cylindrical lens ($f=50$ mm, ThorLabs LJ1821L1-B).

To collimate the output side of the TA, let the seed beam in and leave the TA running at 800 mA. Then collimate the output beam on the vertical direction by adjusting the output lens tubes. We found easier to do this after placing the cylindrical lens roughly about its focal length away from the output side of the chip. This is because as I mentioned earlier the output beam begins to diverge very quickly on the horizontal plane. So if the cylindrical lens is not placed initially then it would be really difficult to see the beam a sufficient distant away from the output side. After correcting the vertical direction using the aspheric lens next thing would be to collimate the beam on the horizontal plane. This can be achieved by moving the cylindrical lens back and forth along the path of the output beam.

Since the back reflections must be prevented at any cost the cylindrical lens should be tilted slightly for the protection of the chip. One should also use a good optical isolator to prevent any further back reflections. In our case we used two Conoptics M712B isolators because they could provide total of ~60 dB isolation. After the Isolators the beam is coupled in to a Single Mode Fiber (SM800-5.6-125 from Thorlabs) using a Fiber Coupler (Thorlabs-PAF-X-11-B). See Figure 15. Coupling the beam in to a single mode fiber is not that easy with these fiber ports. But a clear and a very useful set of instructions are given in the User Manual for these ports by Thorlabs[22] that would help in achieving good coupling efficiency.

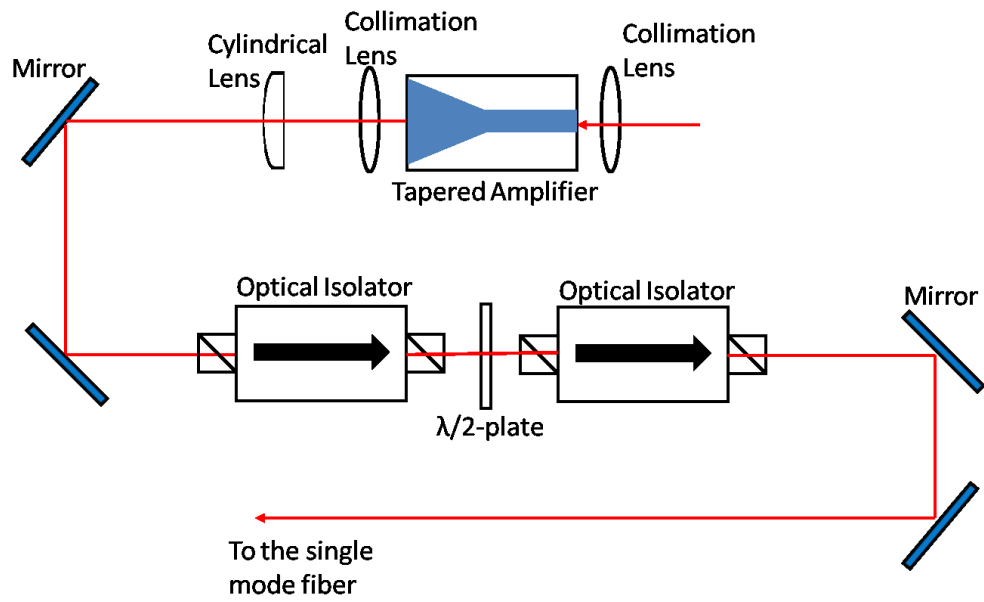


Figure 15: Optical Setup for the TA output

Chapter V - Experimental Data

V.1. TA Characteristics

For the experimental purposes and also for a higher lifetime of the TA chip, it is important to run the TA at its optimum conditions. To find out these conditions one needs to characterize the TA setup. As I have mentioned earlier the life time and the performance of the chip strongly depend on the operating conditions. As an example the output power strongly depends on the coupling efficiency of the seed beam, seed power and also on the TA current. On the other hand, operating the TA at higher currents with lower seed power and poor beam coupling can lower the lifetime of the chip eventually. Due to these reasons we had to characterize the system before we used it in our optical lattice experiments. The following subsections discuss the performance of the chip under various operating conditions.

V.2. Output power vs. Seed power for different TA currents

To see how the output power changes with the seed power, the amplifier was operated at various TA currents and the power of the input seed beam was varied. Our TA chip can be operated with a maximum of 2000mA operating current. But the way our current controllers are designed is that they always put out a little less amount of current even when the current controlling knob is at its maximum position. This was designed as a safety feature. This way we can reduce the risk of damaging the chip by accidentally overloading the input current. Experiment is done for four different current values of the TA. Seed power was varied using a half wave plate and a polarizer while the output power of the TA always is measured after the cylindrical lens (see section IV.3 for the experimental setup on the output of the TA). For most of the optical lattice experiments which we are about to do, a spatially clean beam profile is expected. So the beam was coupled in to a single mode fiber to obtain a Gaussian profile on the output beam (see section IV.3). So on all the graphs depicted below the before and after fiber powers are measured before and after a single mode fiber.

By looking at the graphs (see Figure 16) we see a gradual increase in the output power as both the current and the seed power increase. One of our goals was to find a saturation seed power for the system because driving the TA with excess (unnecessarily) amount of seed power can degrade the TA [16]. Unfortunately in our case the maximum amount of seed power was limited by the output power of the ECDL. So we couldn't find a saturation point for the seed beam. In Fig.16 we see an increasing trend for the output and pre fiber powers as both the current and the seed power increase, but the single mode fiber output tends to level off in all cases. That means the coupling efficiency of the output through the single mode fiber decreases as both the TA current and the seed power increase. The coupling efficiencies for the four TA current values are summarized in a table (see Table 1).

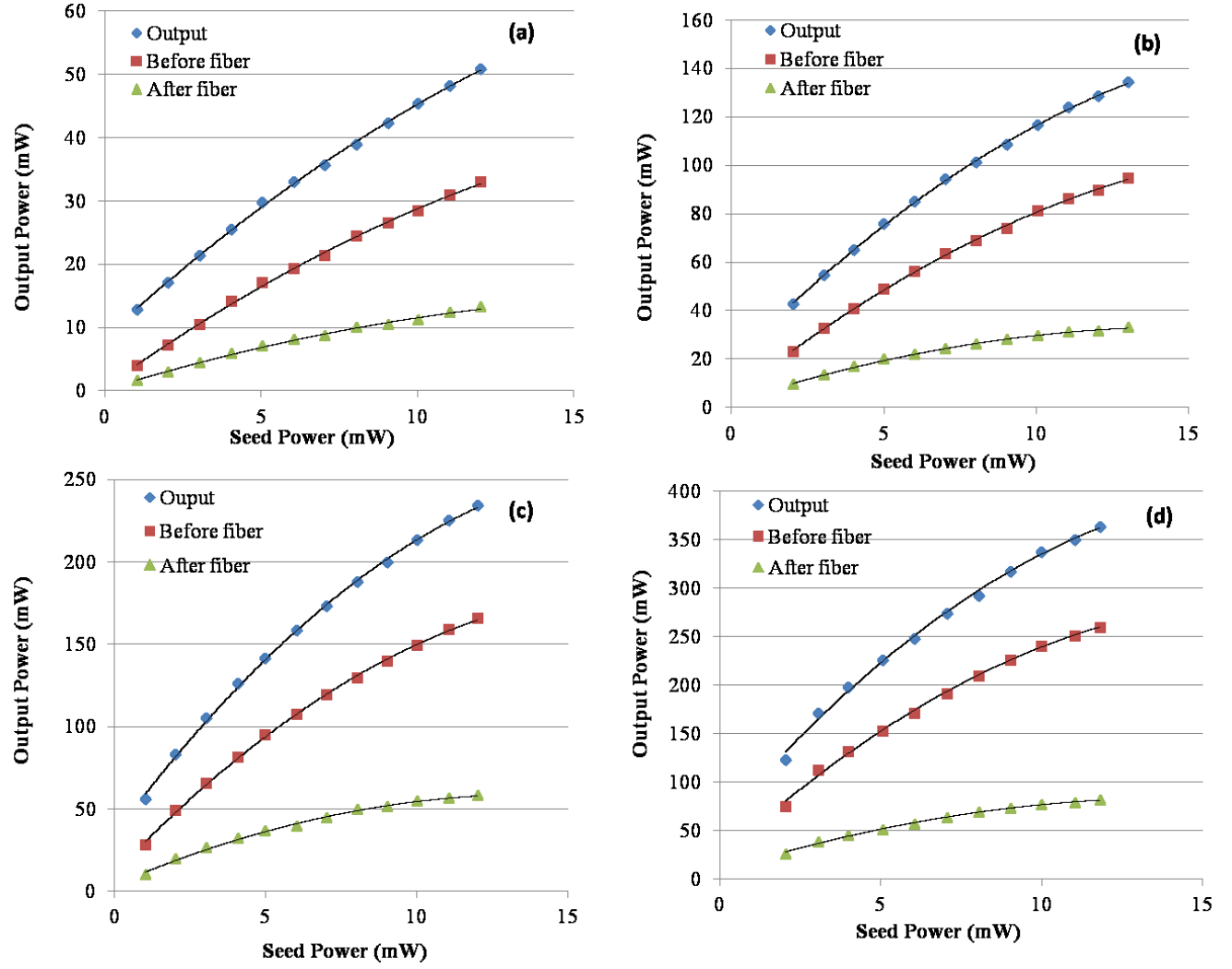


Figure 16: Output power as a function of the seed power at (a) 0.750A (b) 1.00A (c) 1.25A and (d) 1.5A current through the TA

Table 1: Coupling efficiencies for different seed powers at different TA currents

Seed power(mW)	Coupling efficiency (%)			
	0.75A	1A	1.25A	1.5A
2	41.7	41.2	39.7	34.8
3	41.6	41.0	40.2	34.4
4	41.8	41.2	39.5	34.6
5	41.7	40.4	38.7	33.6
6	42.3	39.2	36.8	32.9
7	40.6	38.1	37.2	33.2
8	40.9	38.0	38.1	32.9
9	39.6	38.2	36.8	32.5
10	39.4	36.3	36.6	32.2
11	39.7	36.2	35.6	31.5
12	40.0	35.1	35.1	31.3

Let us first focus on the decrease in coupling efficiency with increasing TA current alone. This agrees with the conventional understanding that as one increases the current through the TA the spatial mode profile of the output beam becomes multimode owing to increased contribution from the ASE compared to the stimulated emission generated by the single-mode seed. This results in poorer coupling efficiency through the single mode fiber at high TA currents, as shown in Figure 17. We have plotted the coupling efficiency for a 12.6mW seed as a function of the TA current. According to the figure it is obvious that the coupling efficiency goes down as the TA current increases.

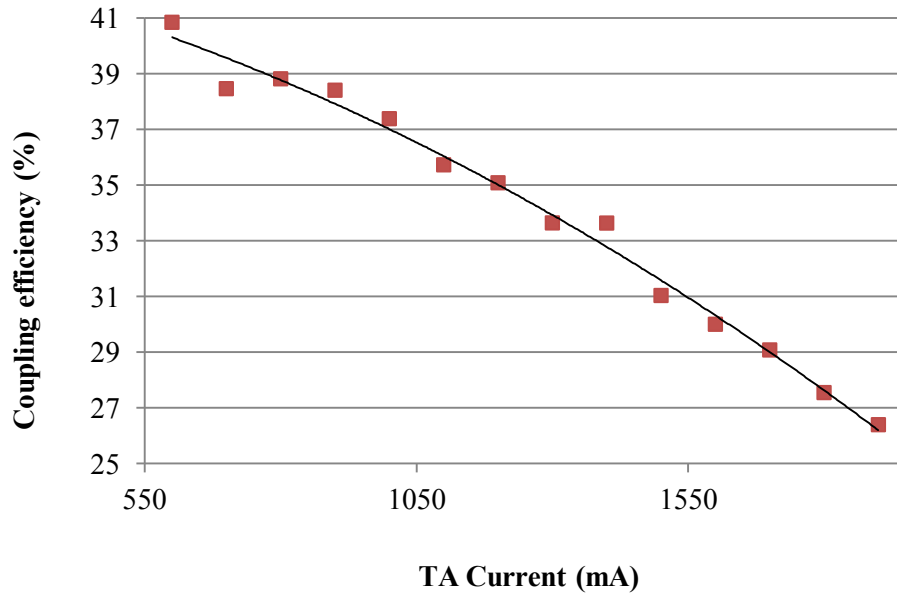


Figure 17: Fiber coupling efficiency versus the TA current for a 12.6 mW seed

Then we plotted the fiber coupling efficiency as a function of the TA current for two different seed powers. Figure 18 shows the for a given TA current, the fiber coupling efficiency is higher for the smaller seed power. Note that this plot is generated using the data shown in Figure 16.

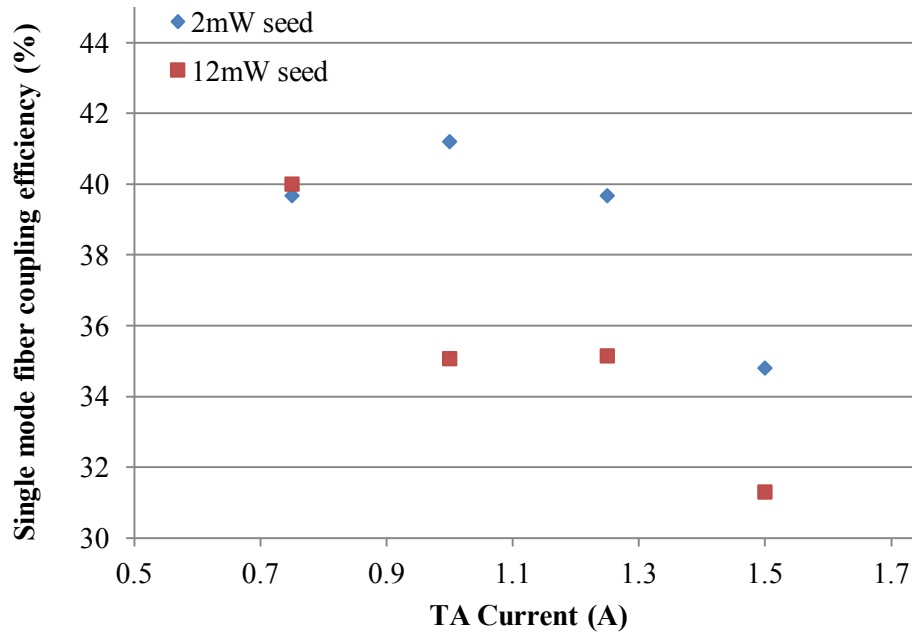


Figure 18: Fiber coupling efficiency versus the TA current for a 2 and 12 mW seed (Data was taken using the TA for the Trapping laser)

Then we did the same test again, but this time with a different TA (TA used for the optical lattice beams) and a different laser. This time we actually did the experiment and measured the fiber coupling efficiency for different TA currents. Again the data is plotted for two different seed powers. TA output and the fiber coupling were optimized at each data point using the coupling mirrors. See Figure 19.

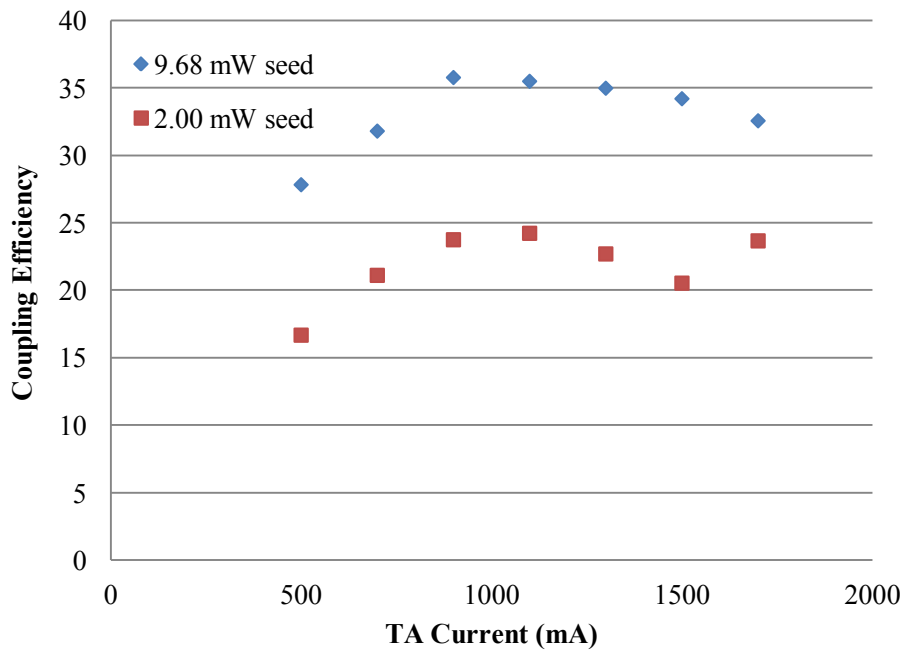


Figure 19: Fiber coupling efficiency for two different seed powers (Data is taken using the Lattice TA)

We observe the same trend for the coupling efficiency for a given seed value except this time higher coupling efficiency is attained for the higher seed power. We could not explain this behavior but we believe the reason for this behavior is a combination of change in the astigmatism and mode profile of the output beam. Astigmatism of the chip strongly depends on the refractive index of the gain medium [21]. Since the refractive index strongly depends on the carrier density and the temperature of the gain medium, distortions of the carrier distribution or the thermal variations have a remarkable influence on the refractive index of the medium [21].

Our data show that for relatively higher values of seed powers ($>9\text{mW}$) and for higher TA currents ($>1\text{A}$) fiber coupling efficiency is in the 30%-40% range. We find that the single mode fiber coupling efficiency, if optimized for a certain TA current, changes by several percent as the TA current is then increased or decreased over a $0.75\text{A} - 1.9\text{A}$ range. As it is mentioned in the previous paragraph we speculate that this change in the fiber coupling efficiency comes mostly from changes in astigmatism of the TA output beam at various TA currents. As a result we believe that one always needs to readjust the collimation of the output beam, after choosing their preferred TA operating current.

V.3. Output power versus the TA current

In the previous section we observed that the overall output of the TA increases with the increase of the TA current. But in the previous case it was difficult to see what kind of a trend that the TA output has when we increase or decrease the TA current. So in this case what we did was that we kept the seed power constant (12.6mW) while increasing the current of the TA and measured the TA output, pre fiber and post fiber powers. See Figure 20. By looking at the graph again we find that the mode profile of the output beam is compromised leading to decreased coupling efficiency through a single mode fiber. Maximum coupling efficiency of 40.8% was observed at 600mA and at 1.9A it dropped down to 26.4%. But one of the interesting facts to notice is that $>100\text{mW}$ power is obtained through the single mode fiber.

We achieved a gain of 16.5dB at 1.9A (for 12.6mW seed power) which compares favorably with the manufacturer's value of 16.5dB for a 20mW seed. The dB-gain value gives us a reasonable parameter to compare TA performance across different seed powers.

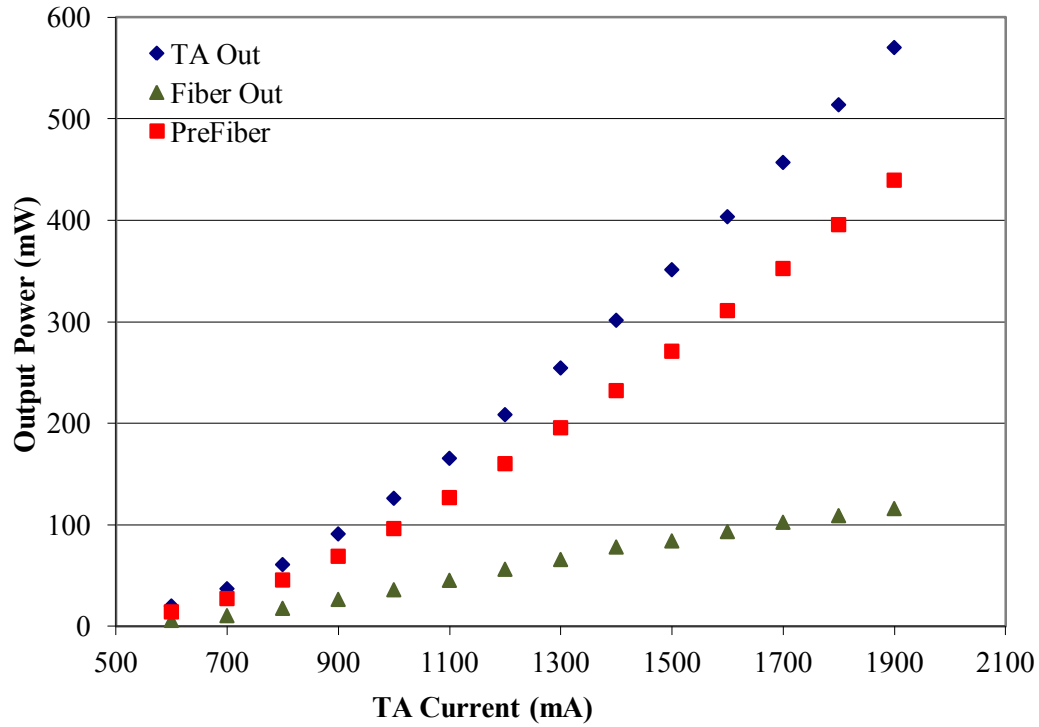


Figure 20: Output power versus TA current for 12.6 mW seed (for the TA used in the trapping laser)

V4. TA output vs. the seed polarization

The output power of the TA strongly depends on the polarization of the seed laser. Just to see how much the polarization affects the output power we change the angle of the half wave plate placed on the input side of the TA and measured the output.

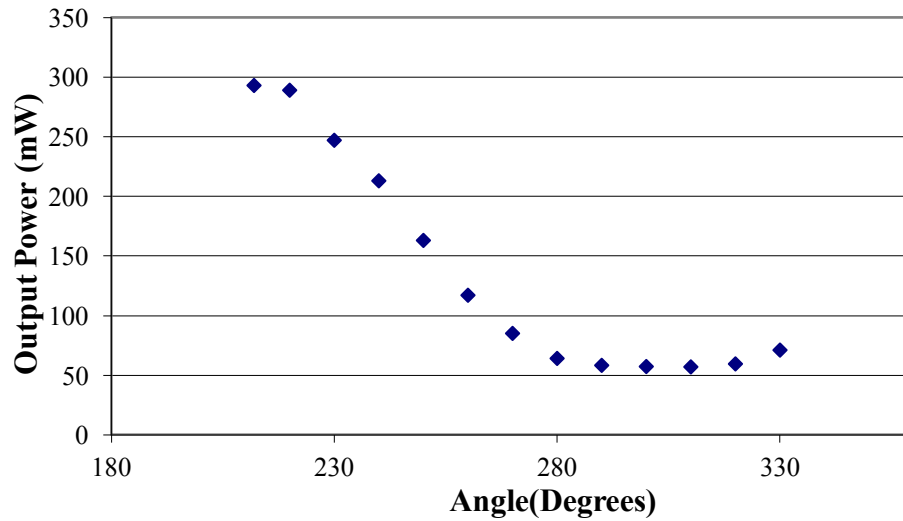


Figure 21: Output Power versus the seed polarization (for the lattice TA setup)

By looking at the graph we see that when the half wave plate gets rotated by $\sim 90^\circ$ the output power goes from a maximum to a minimum value.

Chapter VI- Optical lattice

VI.1. Number measurement

It was important us to measure the number and the optical depth of our ^{85}Rb atom magneto optical trap (MOT) after implementing the TA system for the trapping laser. The number density measurement has been done by observing the fluorescence signal coming out from the cold atom cloud. See Figure 27.

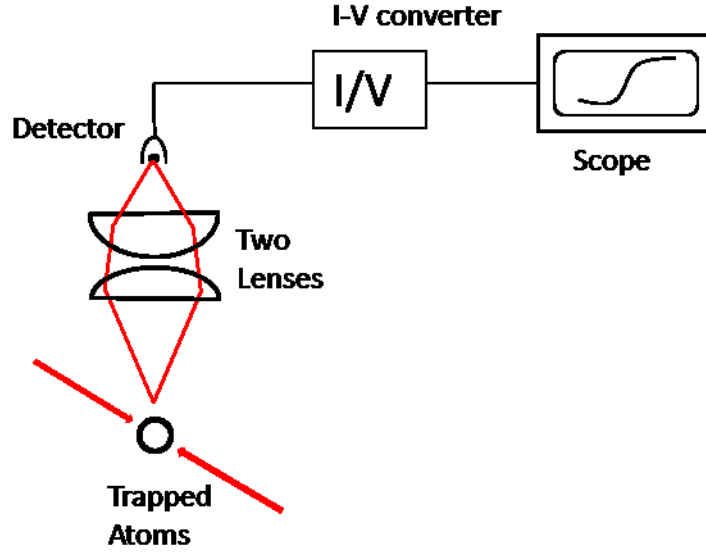


Figure 22: Setup for determining the number of trapped Rb atoms inside the MOT

Inside the MOT, Rb atoms are always absorbing and emitting photons while they are being trapped inside the chamber. The amount of photons emitted by the atom cloud per second depends on the amount of atoms in the excited state (excited state fraction), the photon scattering rate of a single atom and the number of atoms in the trap. It is known that the scattering rate for a single atom γ_p is given by,

$$\gamma_p = \frac{I/I_s \frac{\gamma}{2}}{1 + I/I_s + \left(2\Delta/\gamma\right)^2}$$

where γ is the natural linewidth ^{85}Rb (6MHz), Δ is the detuning of the trapping laser from resonance, I is the intensity felt by the atoms and I_s is the saturation intensity of Rb (1.64 mW/cm²). To measure the intensity felt by the atoms inside the trap one has to take several intensity measurements for the trapping beams before and after the vacuum chamber. Intensity of the trapping beam, I_0 is first measured right after the half wave plate which is used to control the intensities of X-Y and Z trapping beams. We found that only 77.6% of the initial intensity is transferred when the beams reach the vacuum chamber. And the beam goes through the chamber we observed a 14.2% loss of the intensity of beams. That means a 7.1% loss can be observed as the beams reach the midpoint of the chamber. So when the beams see the atoms for the first time

the intensity of the beams going to be $I_0(77.6\%)(92.9\%)$. The beams then go through the other half of the chamber which introduces another 7.1% loss. After exiting the chamber the beam passes through a quarter wave plate and gets retro reflected by a mirror. The total loss of the intensity due to the half wave plate and the mirror is calculated as 2%. So when the beams meet the atoms for the second time the intensity of the beams is going to be $I_0(77.6\%)(92.9\%)(92.9\%)(98\%)(92.9\%)$. So the total intensity, I , felt by the atoms can be calculated as

$$I = I_0(77.6\%)(92.9\%) + I_0(77.6\%)(92.9\%)(92.9\%)(98\%)(92.9\%)$$

$$I = I_0(1.33)$$

At 1000 mA on the TA the total intensity felt by the atoms is 15 mW/cm^2 . So $I/I_s = 9.12 \text{ mW/cm}^2$. The Detuning of the laser is two and a half linewidths (2.5γ). So for a single ^{85}Rb atom,

$$\gamma_p = 0.779 \text{ MHz}$$

In other words a single Rb atom will emit 7.79×10^5 amounts of photons per second. So the total power, P , emitted per second by an atom would be equal to,

$$P = \hbar$$

where $\omega_{3 \rightarrow 4}$ is the atomic frequency of $F=3$ to $F'=4$ transition for ^{85}Rb , $\hbar$ is the Plank's constant $h/2\pi$. So if we calculate the total power per single atom, it would be $1.97 \times 10^{-13} \text{ W/atom}$.

But as it is seen on Figure 22 we are only capable of measuring power emitted by the atom cloud through a small window of our vacuum chamber. So to calculate the amount of power emitted through the window where we have our detectors setup we need to know the solid angle over that particular window. This could be easily calculated by knowing the dimensions of the vacuum chamber. See Figure 23.

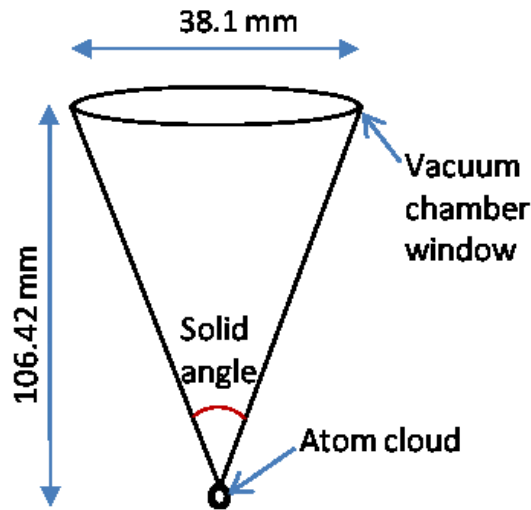


Figure 23: Dimensions for finding the collection efficiency over the window of the vacuum chamber

Solid angle can be calculated as,

$$\text{solid angle} = A/R^2$$

where A is the area of the vacuum chamber window and R is the distance between the window and the center of the vacuum chamber. In our case the solid angle was found to be 0.1 steradians.

Then to calculate the percentage of total power measured over this window we need to divide the solid angle by 4π . Multiplying this percent by P we get the total amount of power emitted by an atom over this window.

$$P_w = (0.1/4\pi) \times 1.97 \times 10^{-13}$$

$$P_w = 1.57 \times 10^{-15} \text{ W/atom}$$

This light is collected by a photodiode which has a response of 0.51 A/W and goes through a current to voltage converter with a gain of 40×10^6 V/A. So by looking at this fluorescence signal we could determine the number of atoms inside the trap, N ,

$$N = \frac{\text{fluorescence Signal(V)}}{1.57 \times 10^{-15} \times 0.51 \times 40 \times 10^6}$$

$$N = \frac{\text{fluorescence Signal(V)}}{3.198 \times 10^{-8}}$$

VI.2. Optical depth measurement

When dealing with magneto optical traps it is important to have an idea about how many that we have trapped per unit volume. Most commonly this is known as the number density. Since we already know how to calculate the number of atoms inside the trap one might say that the density should be the number of atoms divided by the total volume of the atom cloud. But in most cases it is really hard to determine the dimensions of the atom cloud accurately in all three directions. In these cases what people usually do is that they send a probe beam through the atom cloud in the direction where they can find the length of the atom cloud accurately and then measure the absorption of that probe. See Figure 24.

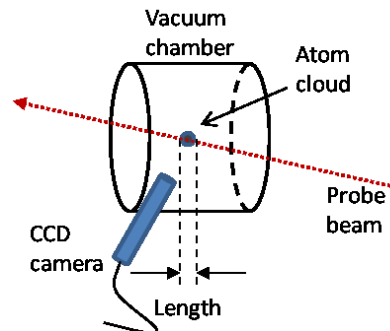


Figure 24: Experimental setup for measuring the size of the atom cloud in the direction of propagation of the probe beam

The probe is on resonance with the D_2 transition of ^{85}Rb ($F=3$ to $F'=4$ hyperfine transition). By taking ratio between the transmitted intensity, I and the incident intensity of the probe beam I_0 on can determine the density of atoms, n inside the trap using the following relationship,

$$I = I_0 e^{-n\sigma l}$$

where l is the length of the atom cloud (see Figure 24) and $\sigma (2.91 \times 10^{-13} \text{ m}^2)$ is the collisional cross section for ^{85}Rb on resonance.

Unfortunately the CCD camera that we were using had a fixed focal length. With this focal length it was really hard to get a focused image of the atom cloud in certain directions (especially in the direction perpendicular to the probe transmission). Because of this reason it became further more difficult to get an accurate measurement of the length of the atom cloud in our setup. With all those reasons, instead of determining the number density we decided to calculate the optical depth of our trapped atom sample. This was more convenient since the optical depth, OD , of a cold atom sample is given by,

$$\begin{aligned} OD &= n\sigma l \\ \Rightarrow \frac{I}{I_0} &= e^{-OD} \\ \Rightarrow OD &= \ln\left(\frac{I_0}{I}\right) \end{aligned}$$

The experimental setup we have used for our optical depth measurement is depicted in Figure 25. The probe beam is derived from the lattice laser. But in this case the laser frequency was scanning around the hyperfine structure of ^{85}Rb . The beam then passes through a neutral density filter and a polarizer. This is important because the intensity of the probe beam needs to be very weak to get an accurate measurement. If the intensity of the beam is too high then the probe beam begins to disturb the atom cloud changing its behavior. Then the beam passes through a plano-convex lens having a focal length of 30 cm before going through the atom cloud. Purpose of this lens is to focus the probe beam through the atom cloud in order to make the whole beam interact with the atom cloud. Lens is chosen in get a smaller focal point compared to the size of the atom cloud (size of the atom cloud can be roughly determined using our CCD camera). Focal spot (the diameter of the focal point) is calculated to be 0.05 mm using the following relationship,

$$W'_0 = \frac{\lambda}{\pi W_0} f$$

where W_0 is the diameter of the initial beam, W'_0 is the diameter of the beam at the focal point, f is the focal length of the lens and λ is the wavelength of the laser.

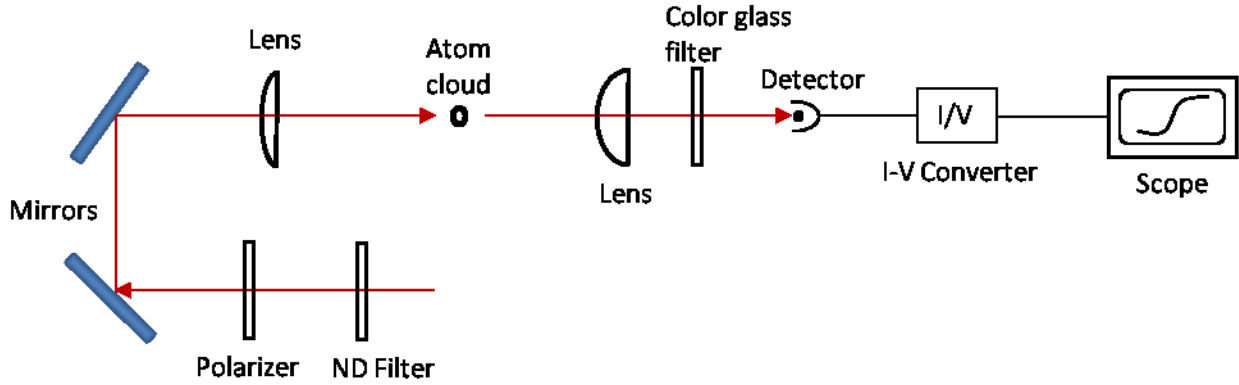


Figure 25: Experimental setup for the optical depth measurement

After the beam travels through the atom cloud it then passes through another plano-convex lens which is used to focus the beam down on to the photo detector. A color glass filter (Newport-model # M3M0810) has been used to filter out any light having a wavelength below 715 nm. This filter prevents the room light falling on to the photo detector and creating any noise on the output signal. Photo detector is then connected to an oscilloscope through a current to voltage converter.

Aligning the beam into the cold atom cloud can be painful because we are trying to hit a 2-3mm ball of atoms using a beam with a waist of less than a 1mm. The method that we used is described below.

In our vacuum chamber the probe beam enters through one of the large windows and exits through the opposite window. We have designed four Plexiglas caps for these windows with markings on their horizontal and vertical axes. See Figure 26.

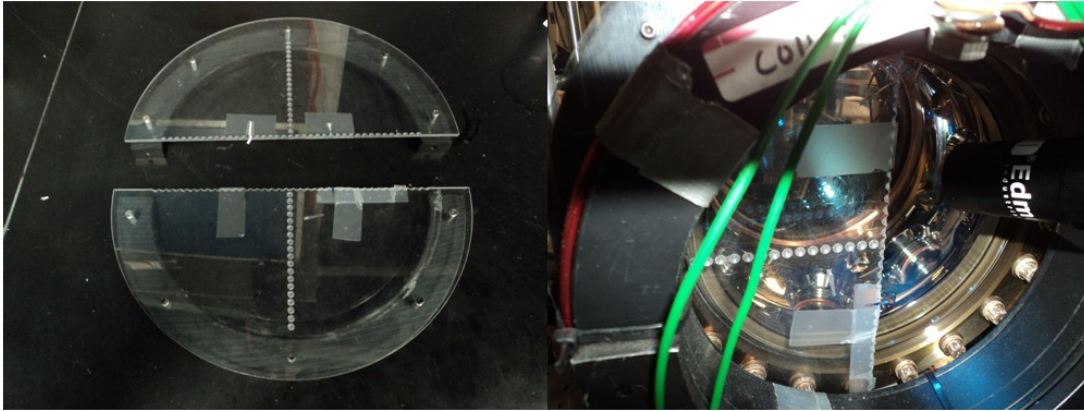


Figure 26: Picture of the window caps

These markings are used to get the initial alignment of the probe beam through the center of the vacuum chamber. When doing this step we removed the 30 cm lens from the probe beam setup. Because when we first align the beam through the atom cloud we wanted the beam to be as big as possible. This makes it easier to find the ball of atoms with the probe beam. After getting the

coarse alignment with the window caps one can move on to the fine alignment using the two mirrors setup before the beam incidents on the ball of atoms. If the resonant probe beam hits the ball of atoms we can observe (on the TV connected to the CCD camera) how the atom cloud gets distracted by the probe beam. Two mirrors were adjusted until we see the maximum pushing effect on the cloud of atoms. At this point we placed the lens back in to the setup. If the beam is focusing down in to the atom cloud boring a hole inside the cloud one can observe a doughnut shape image on the TV screen. Next, one should adjust the polarizer and the neutral density filter to decrease the intensity of the probe beam until the noticeable effect on the atom cloud goes away completely. At this point if we adjust the probe beam perfectly in to the photo detector we can observe the Doppler free hyperfine structure of ^{85}Rb [23]. The probe absorption was then found out with the help of this hyperfine spectrum to calculate the ratio $\frac{I}{I_0}$ [23].

VI.3. Optical lattice

The configuration of the beams that we have used for obtaining the 3D lattice structure is known as *lin* $\perp$ *lin* configuration See Figure 27. The three dimensional periodic lattice structure is generated by four linearly polarized lattice beams [24]. According to the diagram two *y*-polarized beams ($\vec{n}_1 \propto \vec{n}_2$) are propagating on the *xOz* plane. The angle between these two beams, $2\theta_x$, is fixed. From the opposite side two *x*-polarized beams ($\vec{n}_3 \propto \vec{n}_4$) with an angle of $2\theta_y$ between them are propagating on the *yOz* plane. This arrangement produces a periodic modulation of light polarization and intensity which then leads to a periodic modulation of atomic ground state sublevels.[25] Due to these variations in the ground state sublevels the atoms are trapped in potential minima locations where light has pure circular polarization.

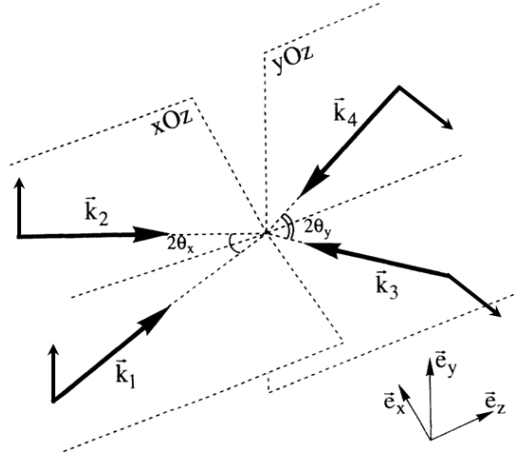


Figure 27: Beam configuration of the 3-D *lin* $\perp$ *lin* optical lattice

One of the main advantages of this configuration is that it does not depend on the phase of the lattice beams [26]. This is important because otherwise one has to phase lock their lattice beams to keep the lattice structure stationary.

VI.4. Detecting the optical lattice

The main idea of this method is to excite the lattice vibrations with a pump and measure the transmission of a probe which passes the lattice structure. This method is widely used in determining the light shift of trapped atoms [26]. The beam configuration used in our lab is given in Figure 28.

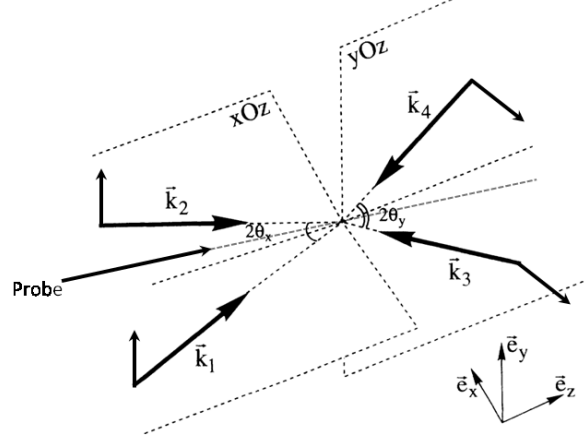


Figure 28: Beam configuration for the pump probe experiment

In this configuration the lattice beams serve as the pump beams and the probe beam comes from the same laser as the beams of the lattice. Probe beam is linearly polarized (either the polarization vector is in the yOz plane or in the xOz plane) and is propagating along xOz plane. Also the probe beam frequency is scanned around the frequency of the pump beams with the help of two acousto optic modulators (AOMs) and a ramp signal provided to one of the AOMs. More details about the experimental setup will be discussed on the upcoming sections of this thesis.

On the experiment we hope to measure the probe transmission as a function of the detuning, δ , between the pump (lattice) and probe beams. The relative detuning between the lattice (pump) and the probe is indicated by,

$$\delta = \omega_{probe} - \omega_{lattice}$$

where ω_{probe} is the probe frequency and $\omega_{lattice}$ is the lattice frequency. We expect to see the peaks due to the Raman resonances between different vibrational energy levels in the potential wells on the optical lattice[26],[25]. See Figure 30. The reason due to two Raman resonances can be understand by looking at the vibrational energy level diagram of a potential well. See Figure 29.

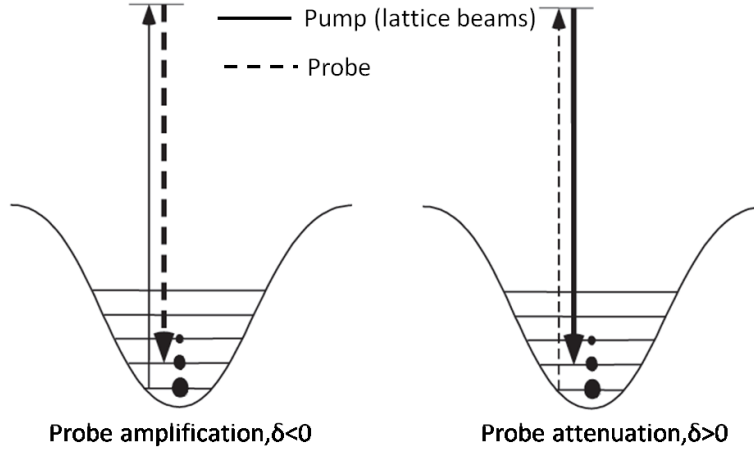


Figure 29: Raman transitions between difference vibrational energy levels of a potential well. The solid circles indicate the population distribution in different energy levels[26].

Assuming the ground state energy levels are more populated than the first excited level, when the probe frequency is less than the pump beam frequency ($\delta < 0$) and the detuning, δ , becomes exactly equal to the spacing between the two vibrational energy levels the process leads to amplification in the probe intensity. When the probe frequency becomes larger than the pump frequency ($\delta > 0$) and that difference becomes exactly equal to the spacing between the vibrational levels one can observe an attenuation in the probe intensity. Figure 30 shows the spectrum for a typical pump probe measurement of a lin –perp- lin optical lattice.

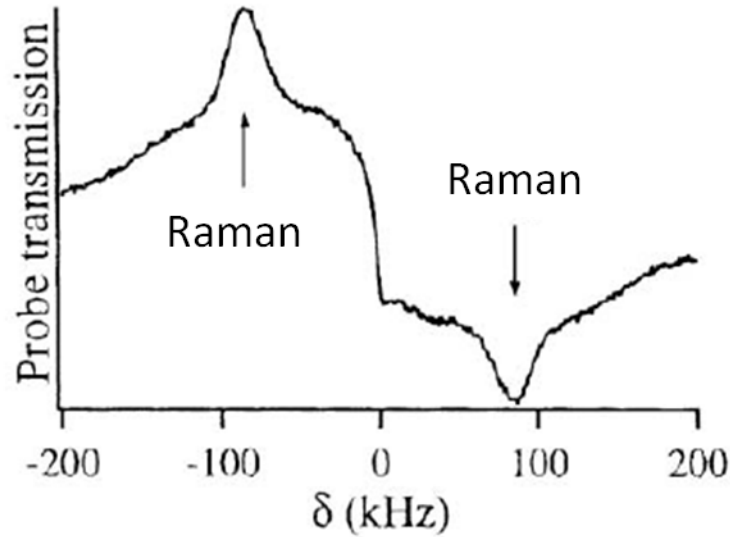


Figure 30: Pump-probe spectrum where the lattice beams serve as the pump beam. Reproduced from the reference [27]

VI.5. Experimental procedure

VI.5.A. Lattice beams

The four lattice beams are derived with the use of an external cavity diode laser and a tapered amplifier. Polarization of the lattice beams are fixed according to the beam configuration given in Figure 27. Intensity of the lattice beams are going to be around $4\text{-}5\text{mW/cm}^2$ while the detuning of the lattice beams is in the range of $9\text{-}10\Gamma$ where Γ is the natural line width of ^{85}Rb [25][26].

VI.5.B. Probe beam

The probe beam must be derived from the same laser which gives us the four lattice beams. This helps to get rid of any phase noises that can arise due to the use of two different sources as pump and probe (reference). The frequency of the lattice beams is kept fixed while the frequency of the probe beam scans around the lattice beam frequency. Scanning is achieved using an AOM. But the problem is if you use a single AOM, it adds an offset frequency shift to the beam and also the scanned beam moves to and fro spatially missing the cold atom cloud. So to keep the beam spatially fixed we needed to use a double pass AOM system. The idea of using two double pass AOM systems is to detune the frequency of the probe by a fixed amount using the first AOM and tune it back to the pump frequency by the use of second AOM.

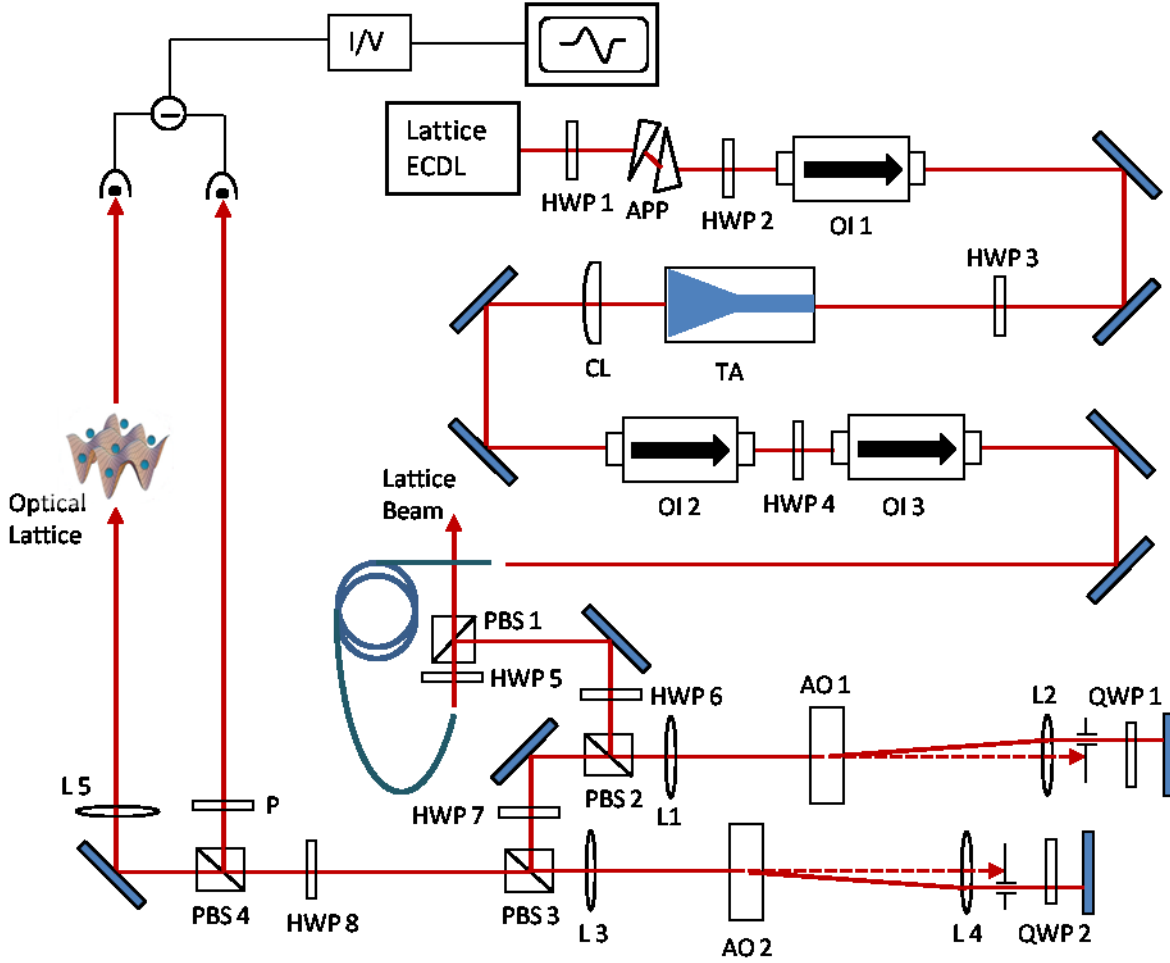


Figure 31: Experimental setup for the pump probe measurement

Experimental setup up till the single mode fiber is described in chapter IV.1 and IV.3. After the single mode fiber a weak portion of the beam is taken off using a half wave plate (HWP 5) and a polarizing beam splitter (PBS 1). This weak beam then goes to two double pass AOM setups which I will be discussing in more details on the following section, while the main beam gets divided in to four lattice beams. After the two double pass setup the probe beam again passes through a half wave plate (HWP 8) and a polarizing beam splitter (PBS 4). The purpose of this beam splitter is to split the probe beam in to two separate beams. One of the beams is focused in to the optical lattice using a 75cm lens (L5) while the other beam (the subtraction beam) passes alongside this beam without passing through the lattice. The subtraction beam is used to cancel out any technical noises present in the lasers. Using the half wave plate (HWP 8) we can control the relative intensity of the two beams and cancel each other out at the photo diodes.

VI.3.C. Double pass AOMs

Acousto optics modulators are used in variety of applications. In our case we use them as frequency shifters. We have two AOMs from Crystal Optics for the double pass setup (crystals: model numbers of 3080-122 and drivers: 1080AF-AEFO-1.0). These AOMs are capable of shifting the frequency of an incoming beam by 50 MHz to 100MHz. According to the diagram (see Figure 31) small portion of the lattice beam has been taken off using a polarizing beam

splitter (PBS 1) and a half wave plate (HWP 5). Since the probe beam needs to be very weak only a small percentage has been taken off. Then the beam goes through a half wave plate (HWP 6) and a polarizing beam splitter (PBS 2). Half wave plate is adjusted in order to get the maximum intensity toward the AOM (AO 1). 300 mm Lens is used to focus the beam through the AOM crystal in order to maximize the transmission. This is important because the crystal has a small aperture compared to the size of the ongoing beam. Spot size the beam at the focal point is calculated as 0.05mm. One has to be careful not to focus the beam too tightly in to the AO because then the beam will not interact with the spatial refractive index variations in the crystal. Our idea was to down shift the frequency of the initial beam (say ' f ') by a fixed amount ' 2δ ' (we get 2δ frequency shift since both the AOMs are double passed) on the first AO (then the frequency becomes ' $f-2\delta$ ') and up shift the frequency by the same amount ' 2δ ' on the second AO (then the frequency becomes ' $f-2\delta+2\delta=f$ '). And then we can provide a ramp signal to the second AO to scan the probe frequency around the lattice beam frequency (' f '). This way we can produce a beam that scans around the original frequency. Figure 32 shows the arrangement of a typical double pass AOM.

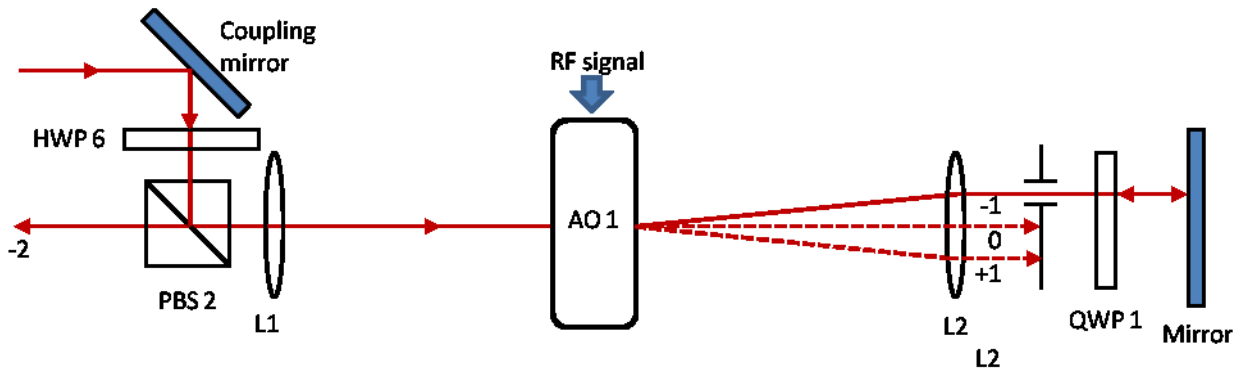


Figure 32: Double pass AOM setup

Acousto optics modulator crystals diffract the light waves using sound waves. Usually an acoustic wave (RF signal) is applied to the AOM crystal by the driver. This wave generates periodic mechanical strains inside the crystal. These strains give rise to a periodic variation in the refractive index of the crystal. So the incident laser beam experiences the Bragg diffraction and also a shift in the frequency of the incoming beam. To get a fixed frequency shift of our first AO (AO 1) we can provide a fixed tuning voltage to the V_T input of our AOM driver. The tuning voltage circuit consists of a voltage divider and an operational amplifier. See Figure 33.

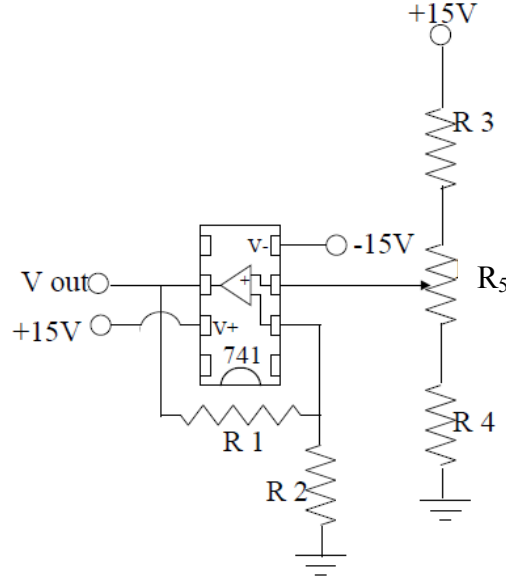


Figure 33: Diagram of the tuning voltage circuit used for the first AOM. $R_1=R_2=19.6\text{k}\Omega$, $R_3=2.2\text{k}\Omega$, $R_4=150\Omega$, R_5 (potentiometer) $=2\text{k}\Omega$ [28]

This circuit is capable of providing a fixed voltage between 1.68V and 14.5V which will give us a 50 MHz to 100 MHz frequency shift. In our case a fixed frequency shift of 80.2 MHz is achieved by applying a 9.68 V to the tuning voltage input of the AOM.

On our first AOM setup first we align the angle of the incoming beam in to the crystal to maximize the transmission on the -1 order (see Figure 32). Diffraction efficiency is measured using optical powers of the 0th (P_0) and the -1(P_{-1}) order of the beam. In our case this number (P_{-1}/P_0) was 63.5%. The diffraction efficiency of the AOM depends on the input angle of the laser beam with respect to the surface of the crystal and also it depends on the RF power level on the driver. When aligning the beam through the AO, first, one should tilt the crystal around its mounted axis to get the maximum brightness on the desired order (in this case it's the -1). Then to get the fine adjustments we have placed a coupling mirror on the path of the incoming beam. To increase the efficiency farther one needs to adjust the RF power level knob on the AOM driver.

The second lens is used to collimate the beam after the AOM and an iris is used to block the undesired orders. After the iris the beam passes through a quarter wave plate and then hits the mirror. This mirror is used to reflect the -1 order back in to the AOM to make it double passed. When reflecting the -1 order back in to the AO crystal to get the initial alignment one could use the iris as a guide. As soon as the reflected beam clears the iris one should be able to see the reflected beam on the output facet of the AO crystal. Then the horizontal and the vertical adjustments on the mirror can be used to make the beam go through the crystal and also overlap it perfectly with the initial beam. When the beam passes through the quarter wave plate twice, the polarization of the double pass beam becomes perpendicular to the polarization of the initial beam. This makes the beam to go through the polarizing beam splitter (PBS 2). Quarter wave plate should be adjusted to get the maximum transmission through the beam splitter. When the -1 order passes the crystal one can again see several diffracted orders. Sometimes it is hard to

identify which one is which. In this case if we opened up the iris completely and turned off the AOM it possible to identify the reflected beam of the 0th order. When maximizing the intensity of the double pass beam, we need to maximize the intensity of the beam which follows exactly the same path as this 0th order beam. See figure below,

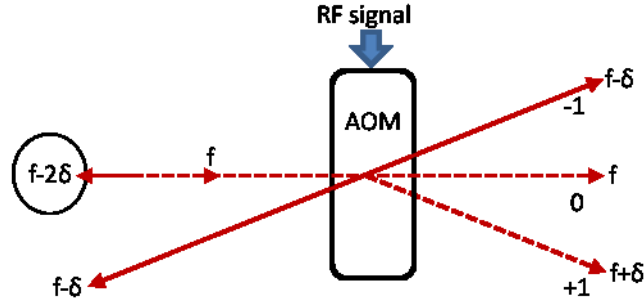


Figure 34: Identifying the correct order of a double pass setup

Diffraction efficiency ($P_{\text{double pass}}/P_0$) of the double pass was calculated as 58.12% (with respect to the 0th order intensity measured right after the AOM on the single pass).

To see if we are getting the desired frequency shift we measured the frequency of the beam before and after the double pass AOM using a spectrum analyzer. The spectrum analyzer consists of a Fabry-Perot interferometer which has a ramping cavity length along with a detector and an amplifier. The output of the spectrum analyzer can be viewed using an oscilloscope where you see peaks corresponding to the wavelength of the incoming light. A detailed description about the spectrum analyzer is given in Nathan Souther's master's thesis [28]. However to use the spectrum analyzer in any kind of an application it needs to be calibrated. This can be done by measuring the period on the oscilloscope which the peaks are repeated over. This value is equal to the free spectral range (FSR=2GHz) of the spectrum analyzer. So by using this we can calibrate the oscilloscope's scale.

When aligning the beams to the spectrum analyzer one should be careful not to increase the intensity of the beam too much. And also the alignment of the beams to the spectrum analyzer becomes crucial because otherwise the peaks become broader and asymmetric. Making the intensity relatively low and using two irises on the path of the beam we could observe narrow and symmetric peaks. Ramp of the spectrum analyzer was adjusted in order to get the maximum spectral range, therefore increasing the accuracy of the data. A detailed description about the observed Fabry-Perot spectrum is given in Nathan Saouter's master's thesis [28].

After the first double pass setup the second double pass AOM was set up. The same tuning voltage was provided in to this second double pass AOM as the one we provided in to the first AOM. But in this case the frequency of the beam was up shifted by ' 2δ '. In this case we could achieve 80.97% diffraction efficiency (P_{+1}/P_0) on the single pass mode and 77.82% ($P_{\text{Double pass}}/P_{+1}$) on the double pass mode. After the second double pass setup the beam was coupled with the initial lattice beam and sent through the spectrum analyzer to see if we had cancelled out the total frequency shift. By doing this we could make sure that our frequency back to the original frequency within 10 MHz because 20 MHz is the minimum resolvable bandwidth of our spectrum analyzer.

To scan the frequency of the probe beam around the pump frequency (lattice beam frequency) we provided a ramp signal to the tuning circuit of the second double pass AOM. The circuit diagram for this tuning voltage circuit looks the same as the tuning voltage circuit that we used for the first AOM except this one has an external input which we can connect to a function generator to scan the tuning voltage around a certain offset value. See Figure 35.

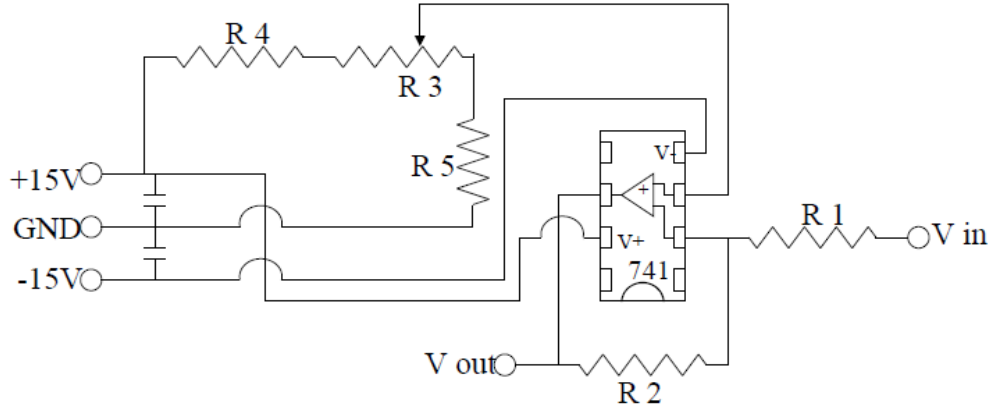


Figure 35: Diagram of the tuning voltage circuit used for the second AOM. $R_1=R_2=33\text{k}\Omega$, $R_4=22\text{k}\Omega$, $R_3=10\text{k}\Omega$ (potentiometer), $R_5=51\text{k}\Omega$ [28]

The circuit provides a tuning range of 3.8V which goes from 8.0V to 11.8V. This corresponds to a frequency range which goes from 76.8MHz to 94.16MHz. In our case we have set the offset voltage to 9.68V and a function generator output with a triangular burst is provided to scan the probe frequency around the pump frequency.

Chapter VII – Conclusion

In conclusion we have presented a user-manual style description of how to build a cost effective, fail-safe TA system suitable for a variety of optics experiments. Instructions are provided on how to operate TA systems safely increasing their lifetime and the output power. Two tapered amplifier systems have been successfully installed in our lab for our optical lattice experiments. We worked hard until the last moment to obtain the results for the pump-probe spectroscopic results. But the lack of time and various experimental obstacles prevented us from getting in to our final results. As a result no data reading our first lattice experiment (pump-probe spectroscopy) are presented in this thesis.

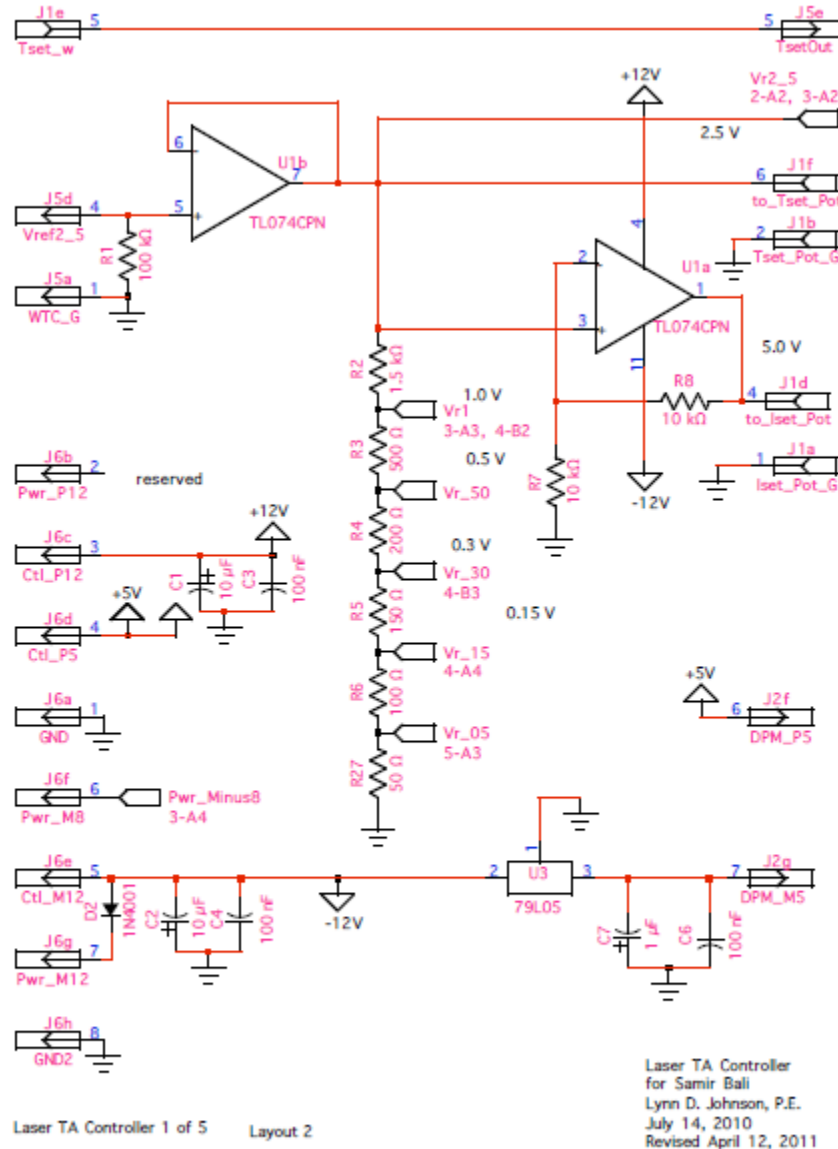
Appendix

A. Circuit Diagrams

Below is an overview of the TA Laser & Temperature control system. This overview is based on previous design notes made by predecessor, Lynn Johnson, who did the design work on the three units used in Dr. Samir Bali's lab.

A.1. Temperature and current controllers

Diagram contains the power connections, voltage reference buffers, and dividers. The reference is a 2.5-volt reference from the temperature regulator device.



The diagram illustrates the internal circuitry of the Tact2 module, which consists of two identical signal processing channels for Tact and Tset inputs.

Top Channel (Tact):

- Input:** Tact signal from connector J1r (pin 18).
- Gain Stage (U2a):** A TL074CPN op-amp configured as a non-inverting amplifier. The input is connected to pin 2. The feedback path from pin 1 to pin 4 contains a 22 kΩ resistor (R9). The non-inverting input (pin 3) is connected to a voltage divider consisting of an 841 Ω resistor and a 10 kΩ resistor (R14) connected to the +12V supply. The output of U2a is connected to pin 9.
- Offset Stage (U2c):** A second TL074CPN op-amp configured as a voltage follower. Its input (pin 9) is connected to the output of U2a. Its output (pin 8) is connected to pin 10.
- Offset Adjustment:** A 10 kΩ resistor (R16) connects pin 10 to the offset input (pin 1) of U2c. A 10 kΩ resistor (R13) connects the offset input to the -12V supply. A 10 kΩ resistor (R17) connects the offset input to the +12V supply.
- Output:** The final output is connected to J2b (pin 2).

Bottom Channel (Tset):

- Input:** Tset signal from connector J1p (pin 16).
- Gain Stage (U2b):** A TL074CPN op-amp configured as a non-inverting amplifier. The input is connected to pin 6. The feedback path from pin 7 to pin 4 contains a 22 kΩ resistor (R10). The non-inverting input (pin 5) is connected to a voltage divider consisting of an 841 Ω resistor and a 10 kΩ resistor (R14) connected to the +12V supply. The output of U2b is connected to pin 13.
- Offset Stage (U2d):** A second TL074CPN op-amp configured as a voltage follower. Its input (pin 13) is connected to the output of U2b. Its output (pin 14) is connected to pin 12.
- Offset Adjustment:** A 10 kΩ resistor (R15) connects pin 12 to the offset input (pin 1) of U2d. A 10 kΩ resistor (R13) connects the offset input to the -12V supply. A 10 kΩ resistor (R17) connects the offset input to the +12V supply.
- Output:** The final output is connected to J2c (pin 3).

Power Supply: The circuit is powered by +12V and -12V rails. The +12V supply is connected to pins 1 and 15. The -12V supply is connected to pins 11 and 14.

Schematic contains the laser diode current monitor circuit in addition to the comparators, which drive the temperature indicator LEDs. This system is referred to as a “Goldilocks Monitor” because it drives the following LEDs: “Too Hot”, “Too Cold”, and “Just Right”.

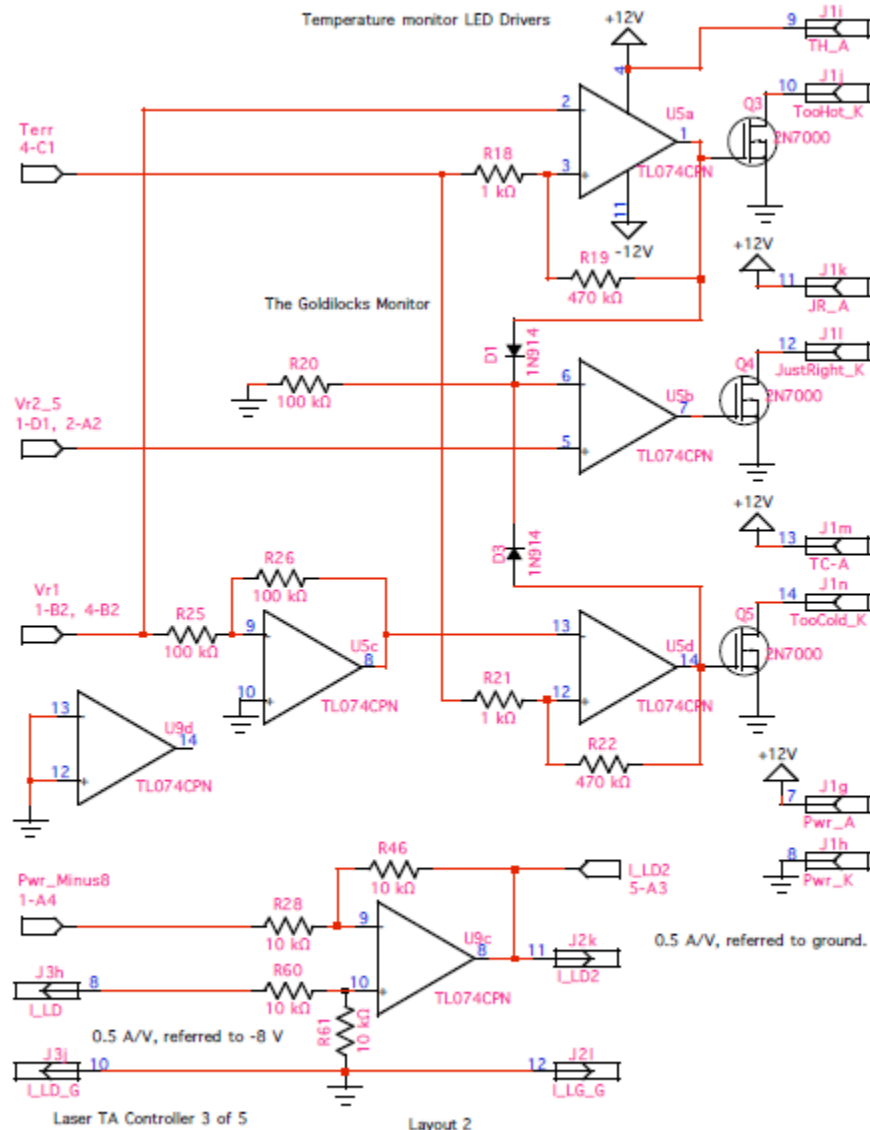
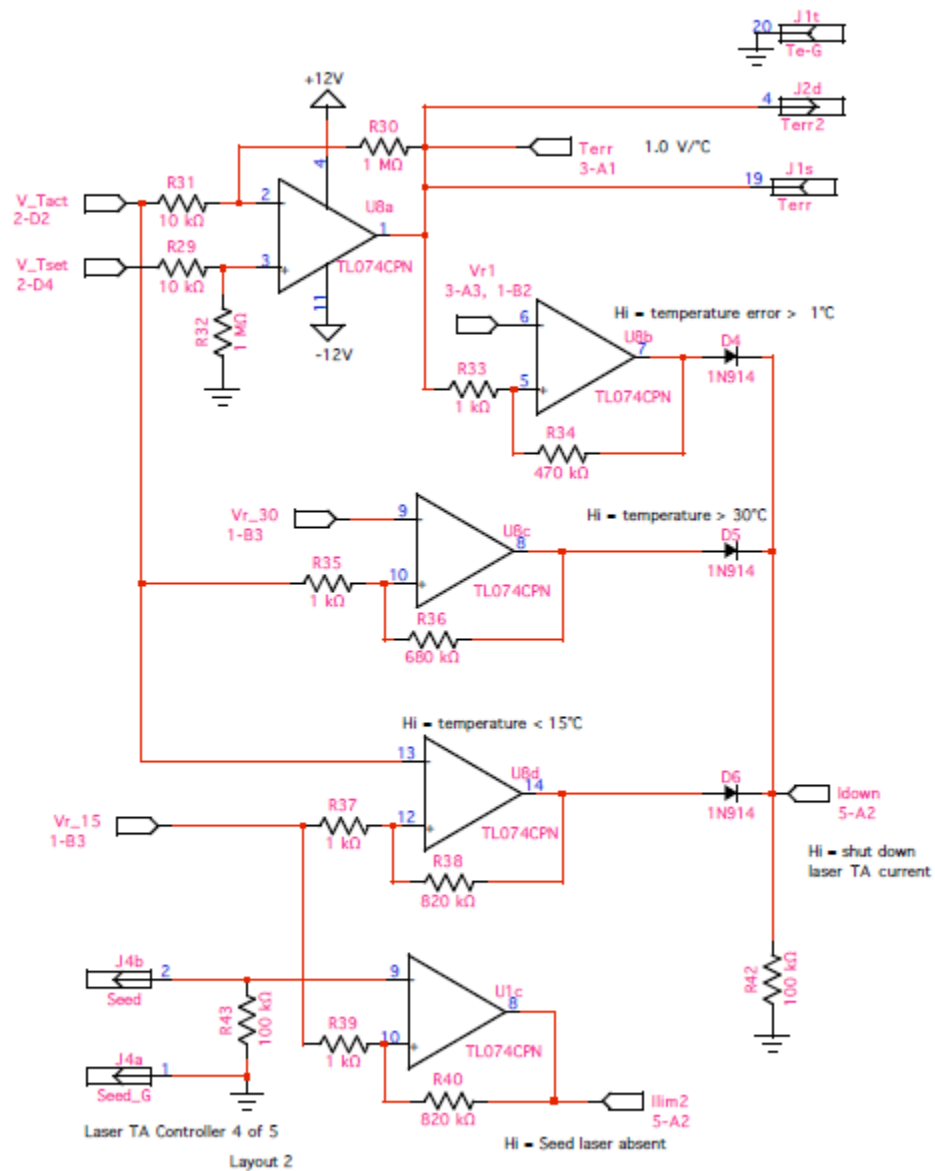
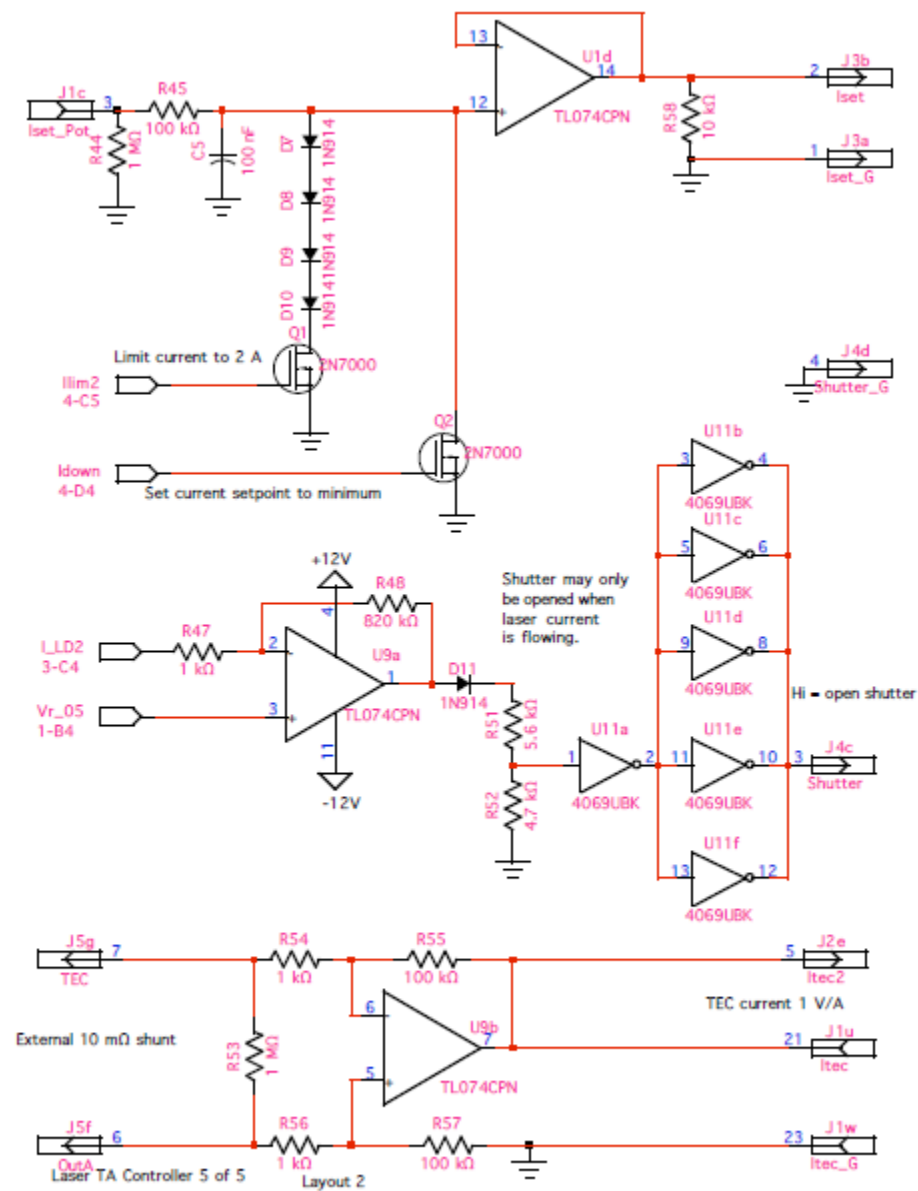


Diagram contains the temperature error circuit in addition to the laser current limiting comparators.

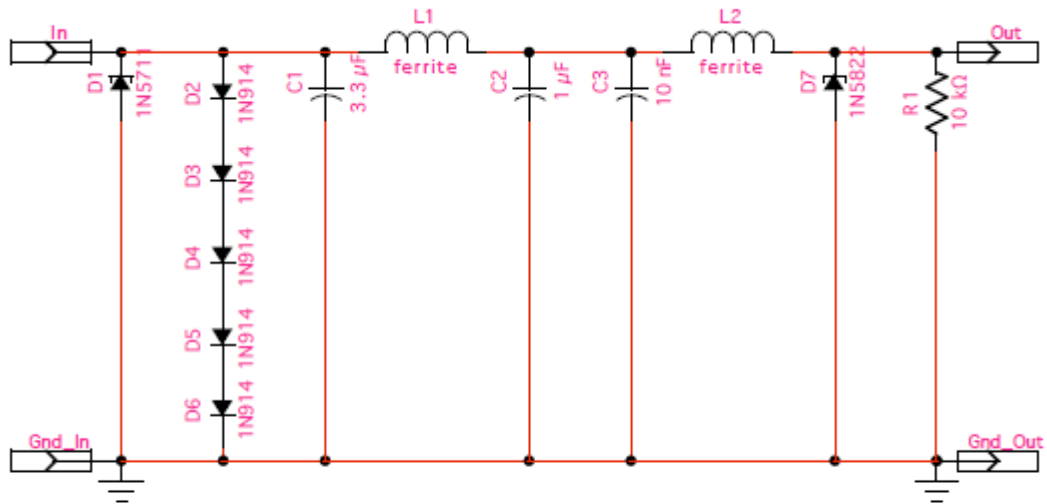


Contains the current limit circuit, the shutter driver (to external control, via +5 Vdc), and the current monitor for the TEC module used.



A.2. Protection circuit

The purpose of this circuit is to further reduce noise and to aid in limiting the amount of current and fluctuations in signal that are delivered to the TA chip.



Laser Diode Protection
Lynn D. Johnson, P.E.
July 29, 2010

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